

Cryoseismology

An aerial photograph of a glacier flowing through a mountainous landscape. The glacier is a light blue/white color, contrasting with the dark, rocky terrain. The flow lines of the glacier are visible, showing its movement from the upper left towards the lower right. The surrounding mountains are dark and rugged, with some snow patches. The sky is a clear, pale blue.

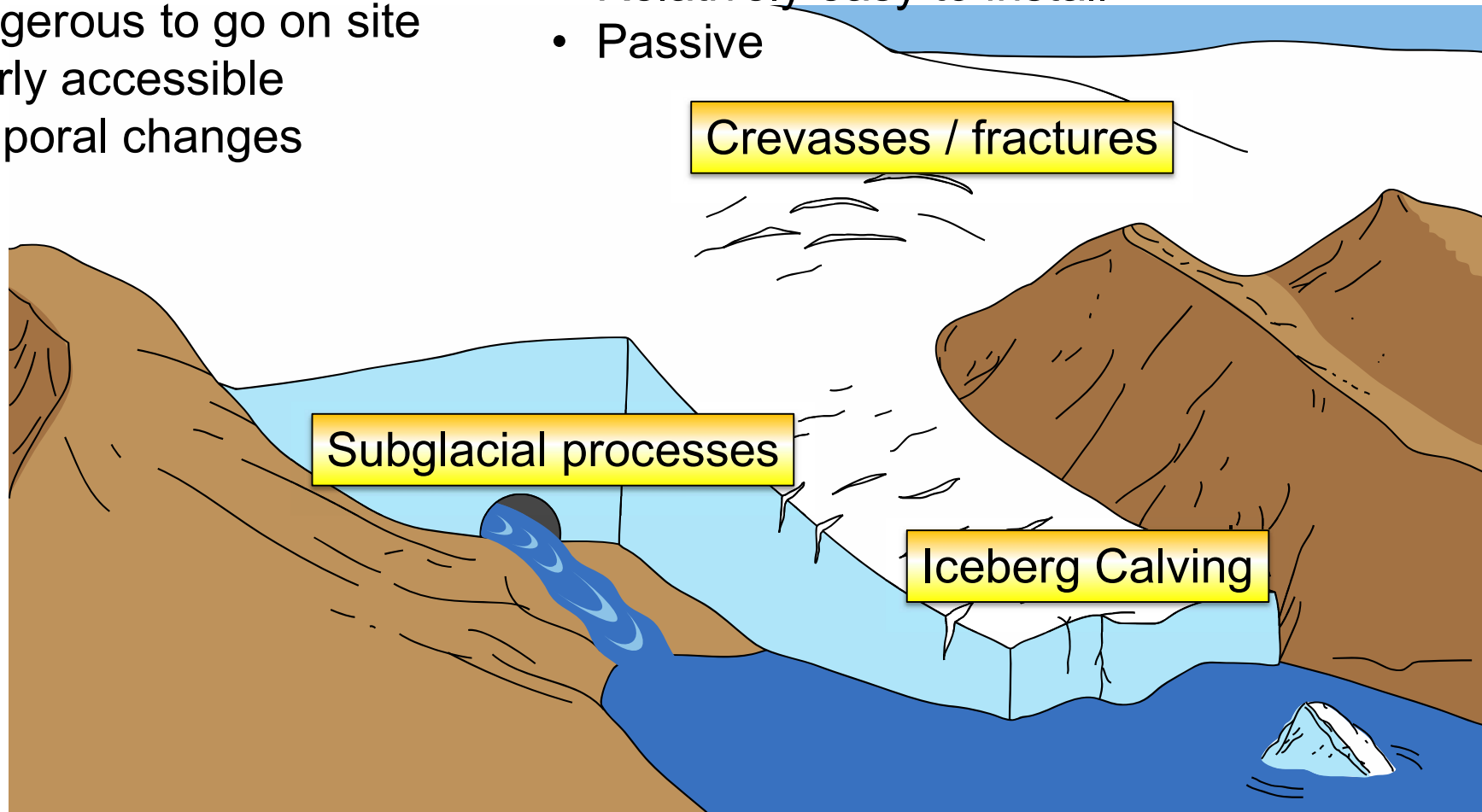
Fabian Walter
Swiss Federal Institute for Forest,
Snow and Landscape Research WSL

Glacier features

- Dangerous to go on site
- Poorly accessible
- Temporal changes

Advantages seismology

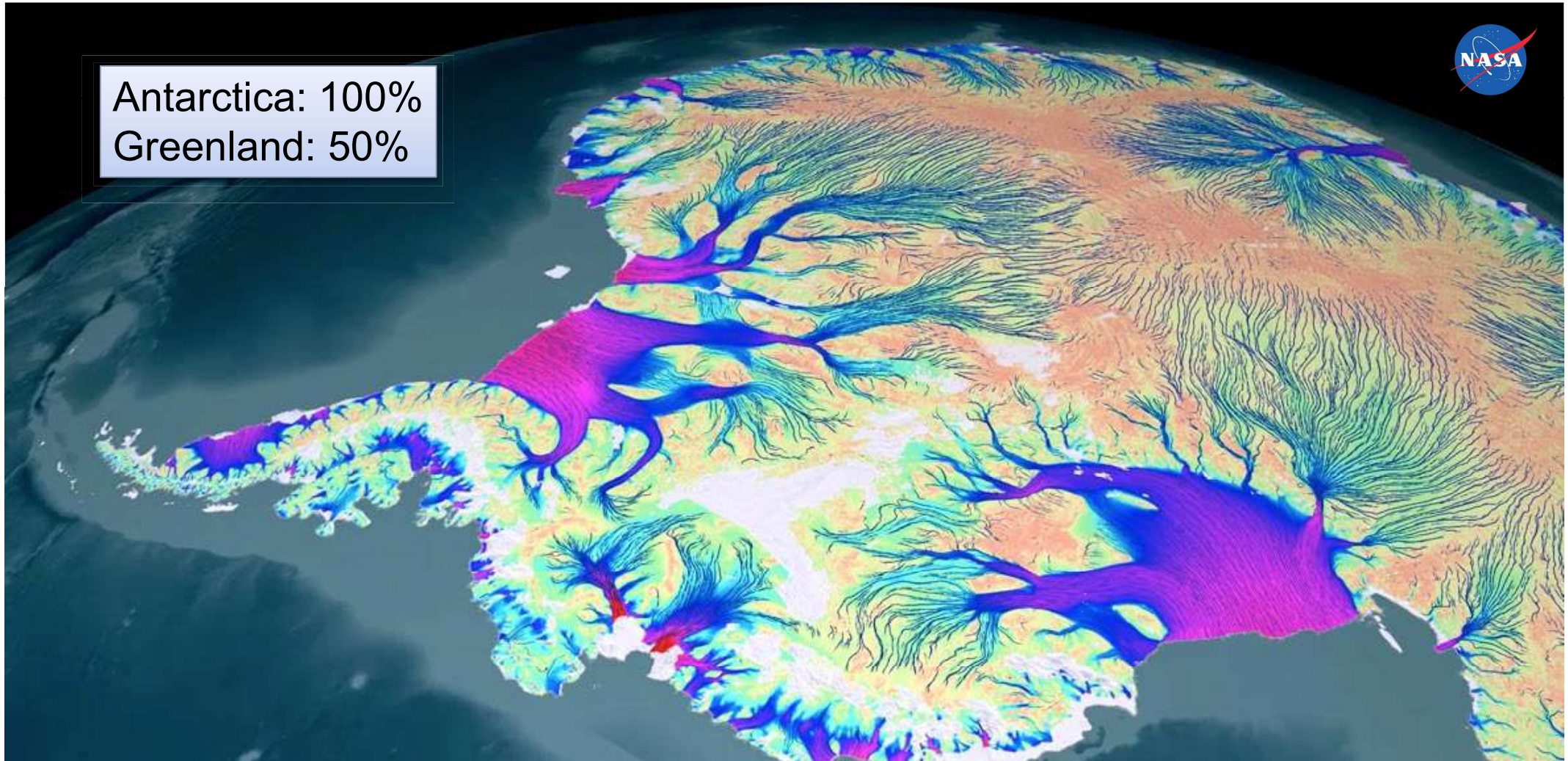
- Remote
- Relatively easy to install
- Passive

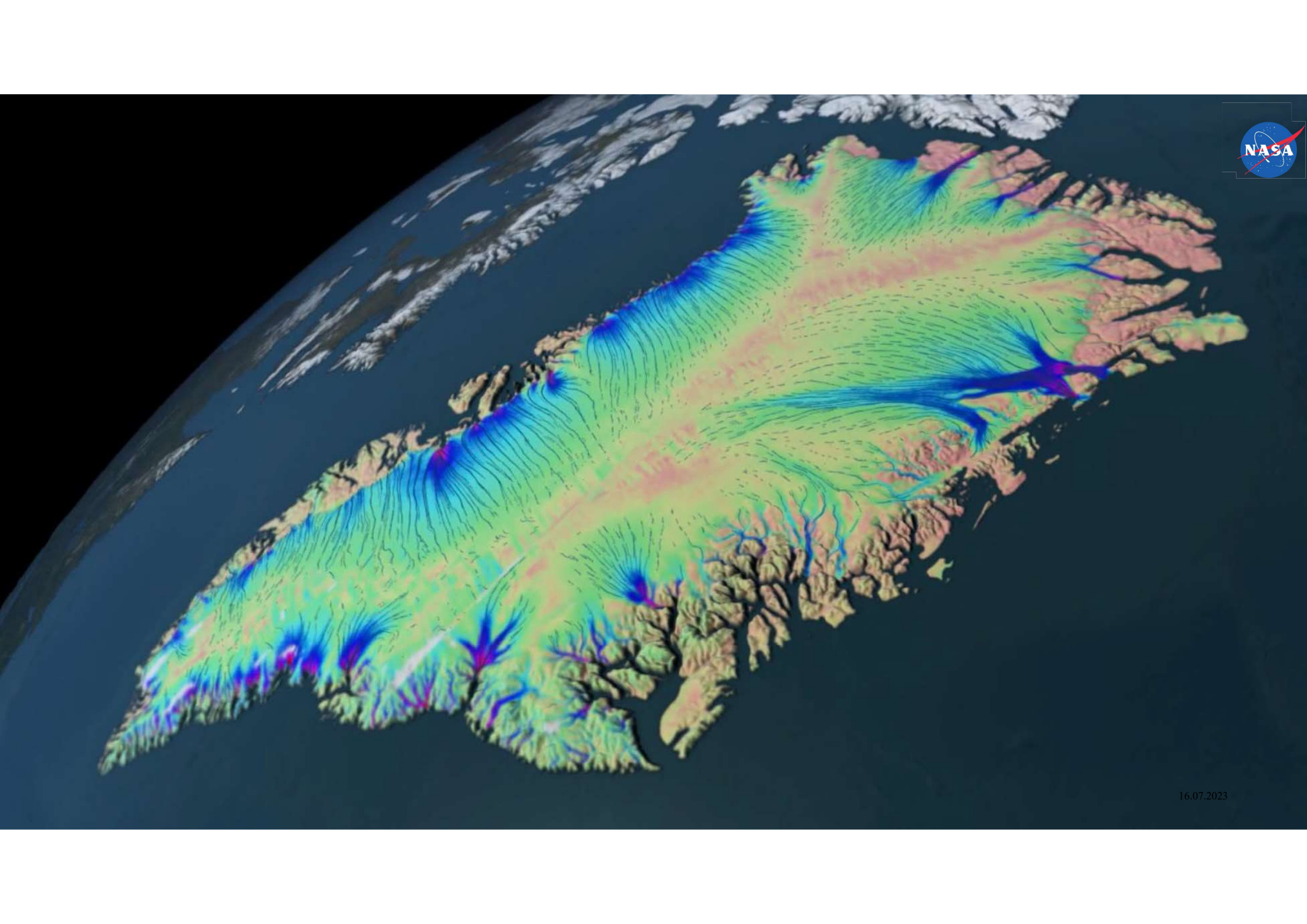


Basal Sliding: Dynamic Mass Loss vs. Surface Melt



Antarctica: 100%
Greenland: 50%

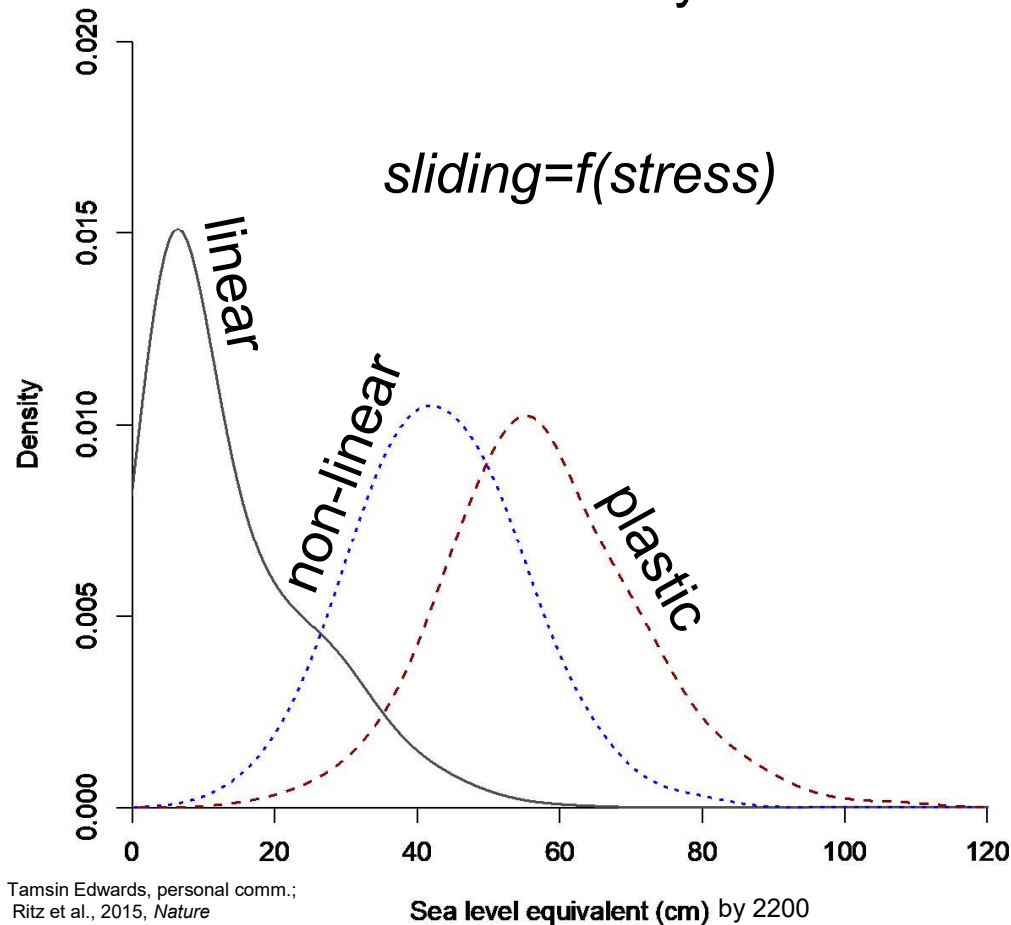




16.07.2023

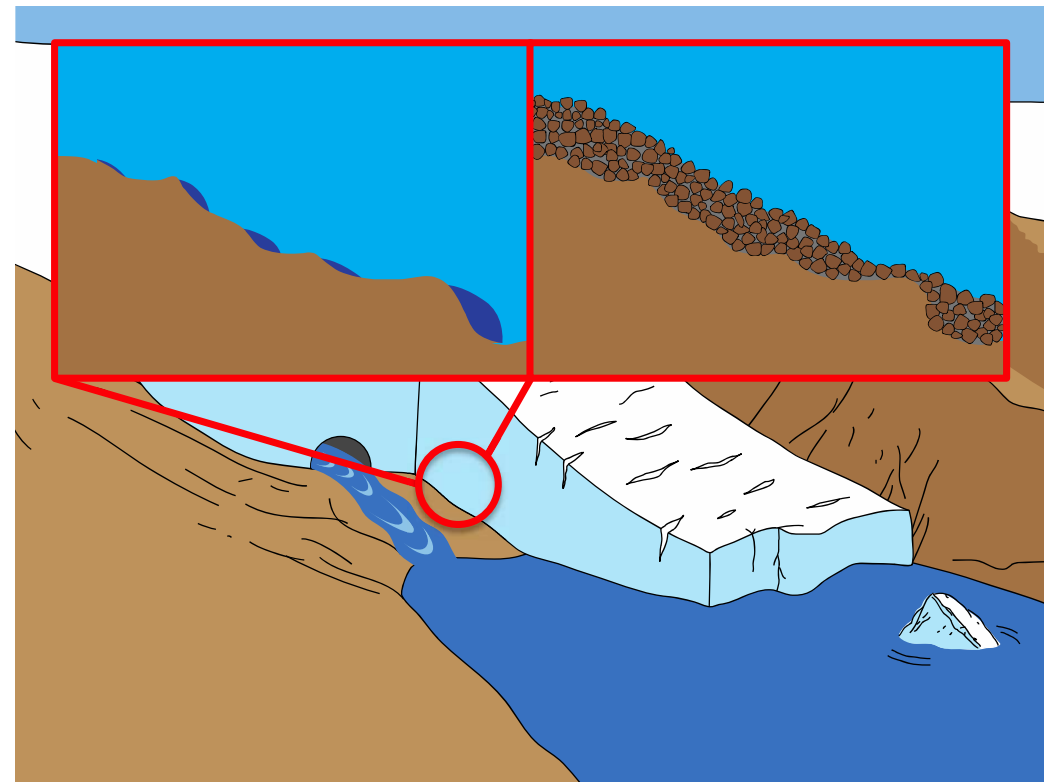
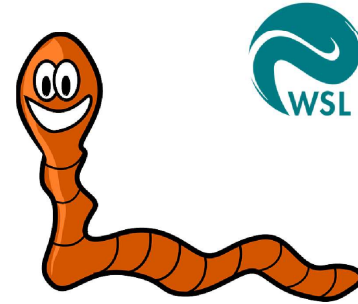
Basal Sliding

Antarctic mass loss by 2200

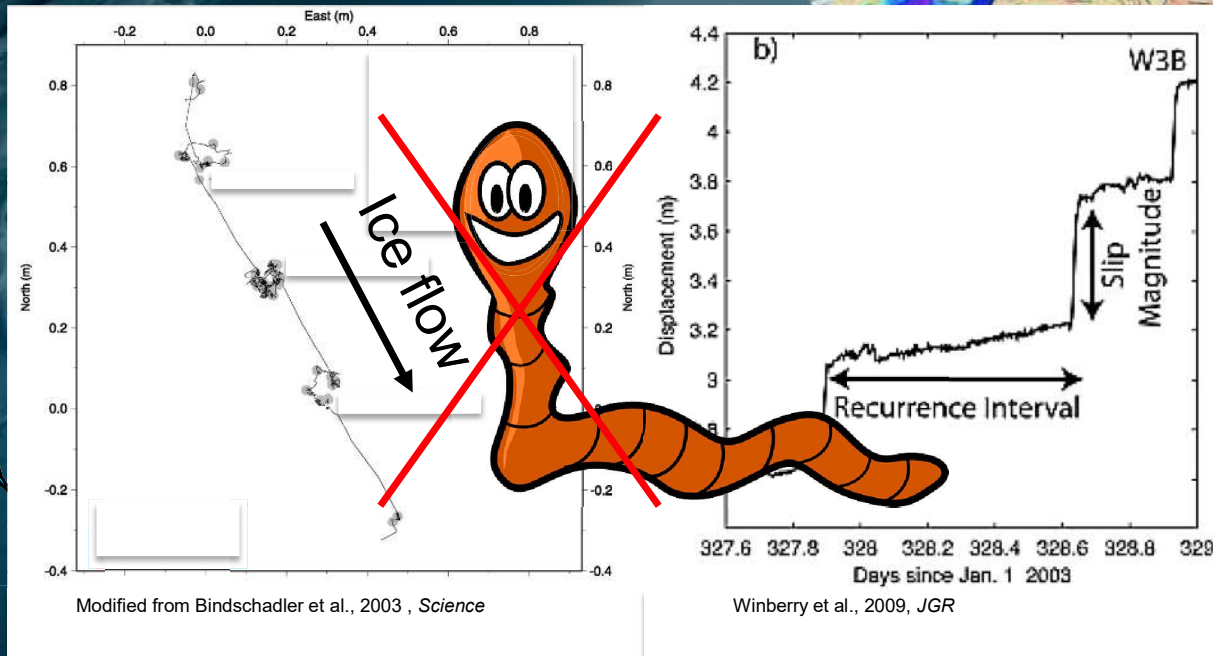


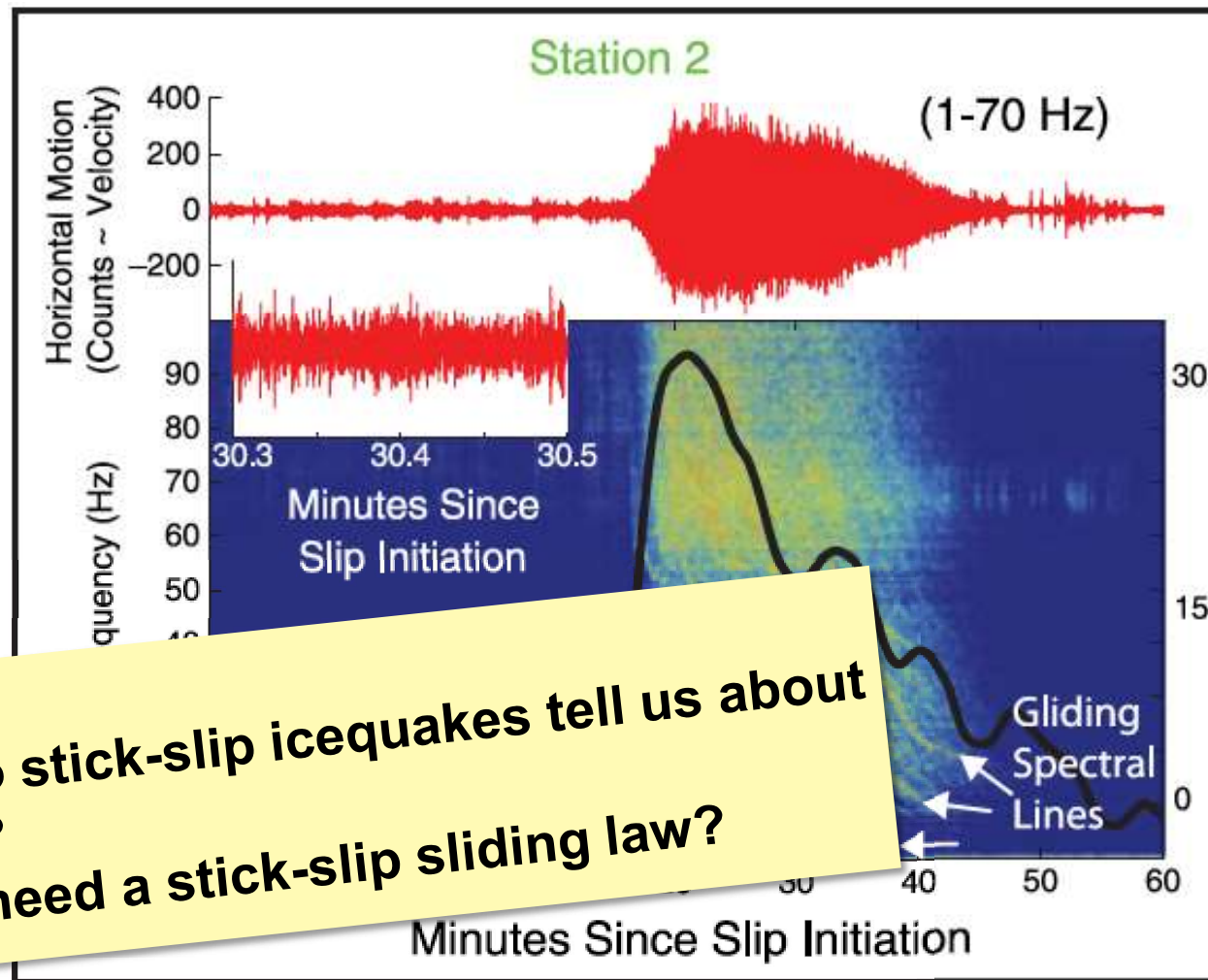
Sliding mechanisms

- Regelation
- Deformation (ice, till)
- **Water pressure (!)**



Antarctica: Basal Sliding



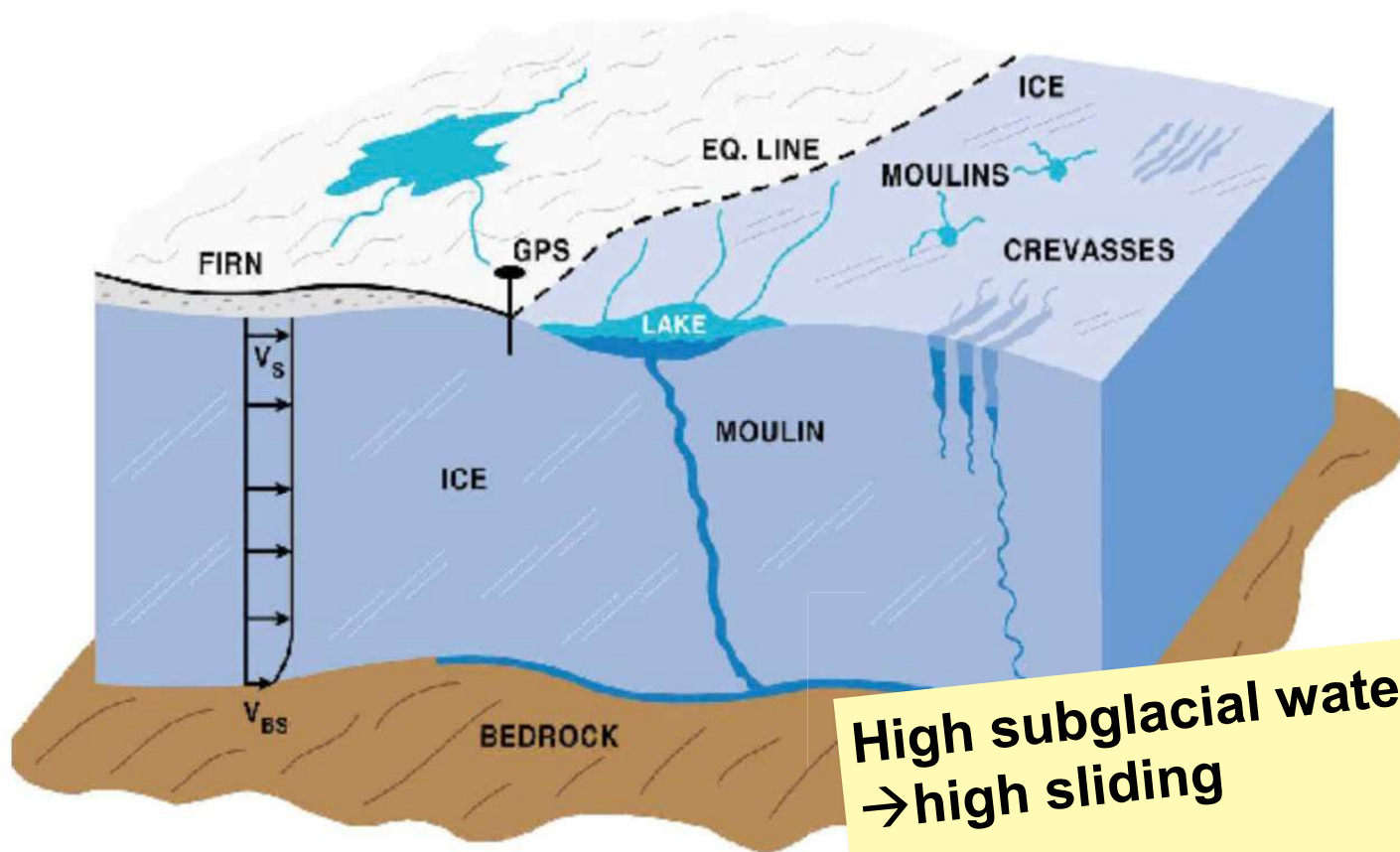


So:

- What do stick-slip icequakes tell us about sliding?
- Do we need a stick-slip sliding law?

Winberry et al., 2011, *Earth and Planetary Science Letters*
 Winberry et al., 2013, *GRL*

Glacier Hydraulics

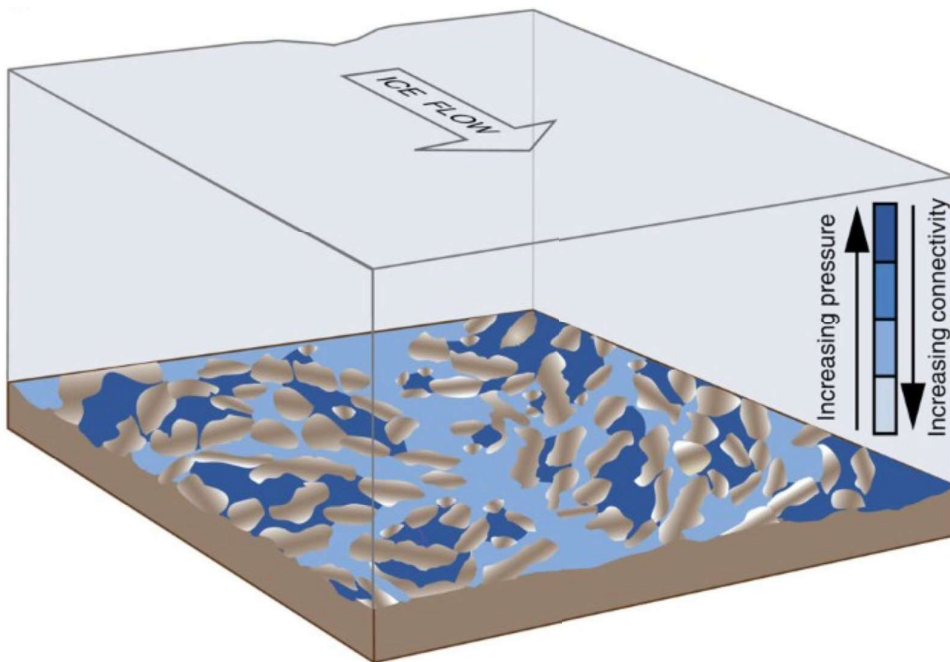


**High subglacial water pressures
→ high sliding**

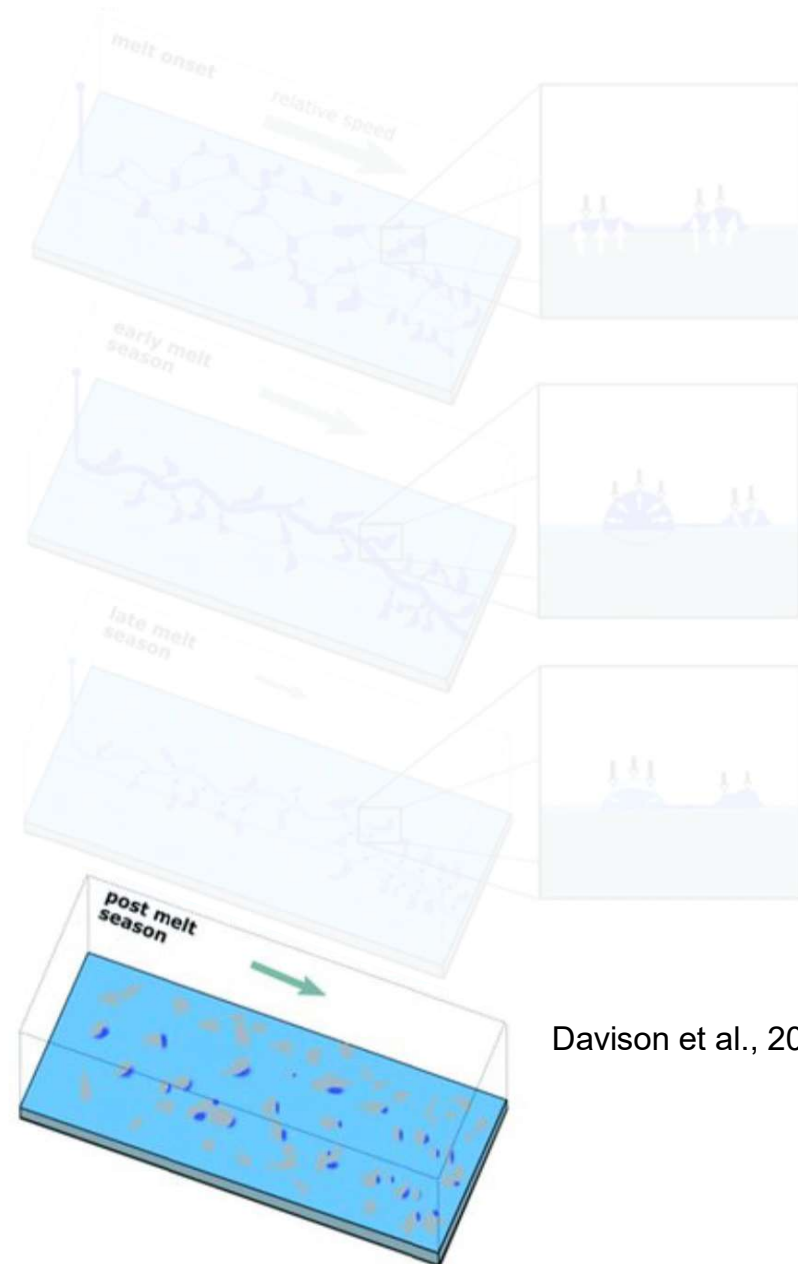
**Efficient subglacial channels
→ low water pressures**

Zwally et al., 2002, *Science*

The Subglacial Drainage System



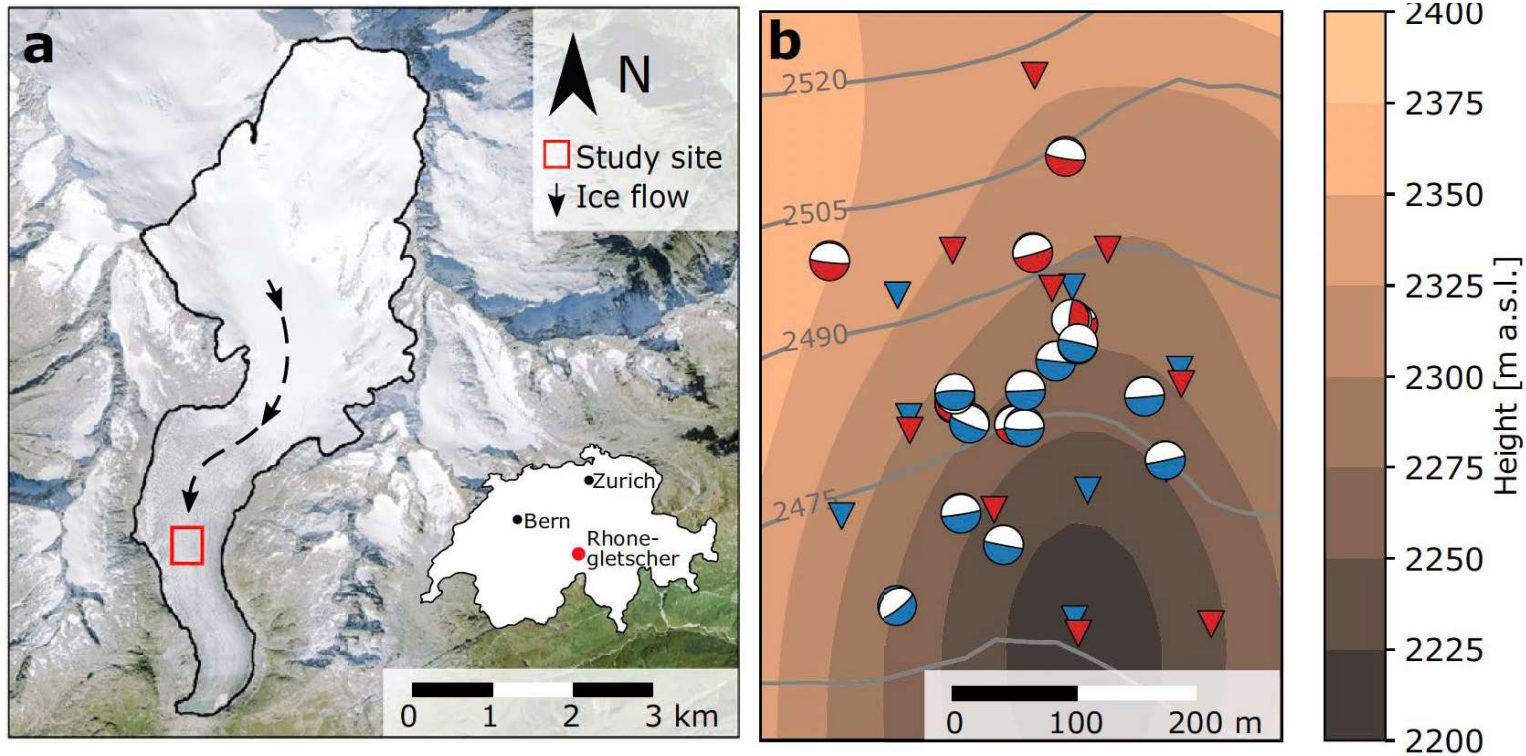
Hoffmann et al., 2016, *NatComm*



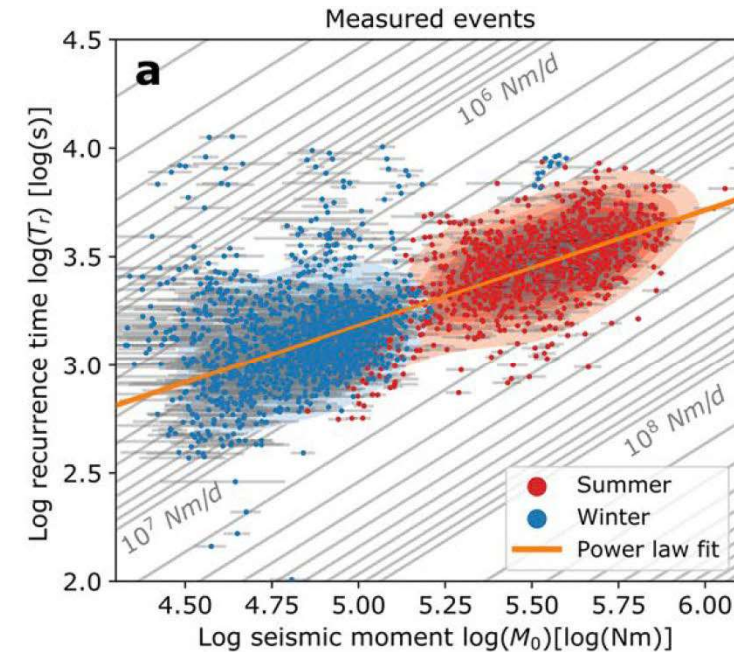
Davison et al., 2019, *Front. Geosci.*

Stick-Slip Asperities in Summer and Winter

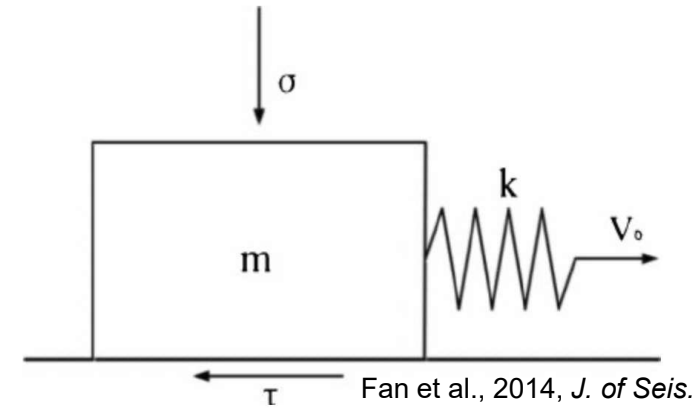
Repeaters - Asperities



Summer vs. Winter Activity



- Larger stick-slip in summer
- Higher moment rate
- Longer inter-event times



Velocity weakening: $b > a$

$$\mu = \mu_0 + a \ln(v/v_0) + b \ln(v_0\theta/L)$$

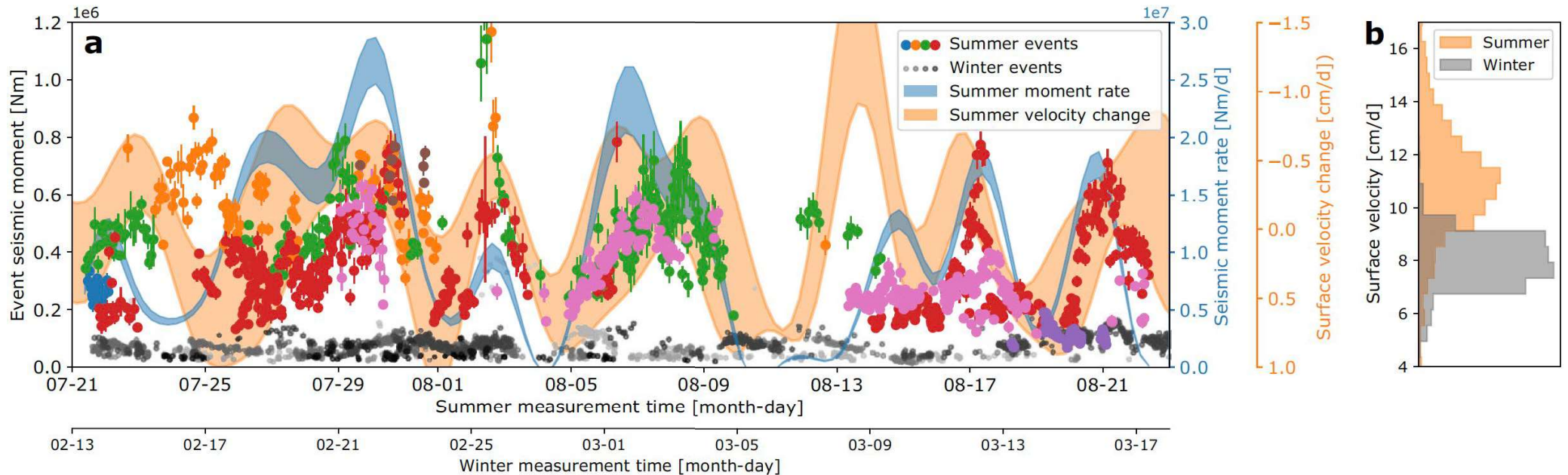
μ_0, v_0 : steady state variables

a, b and L : material variables

θ : state variable (healing, strengthening)

Effective normal stress

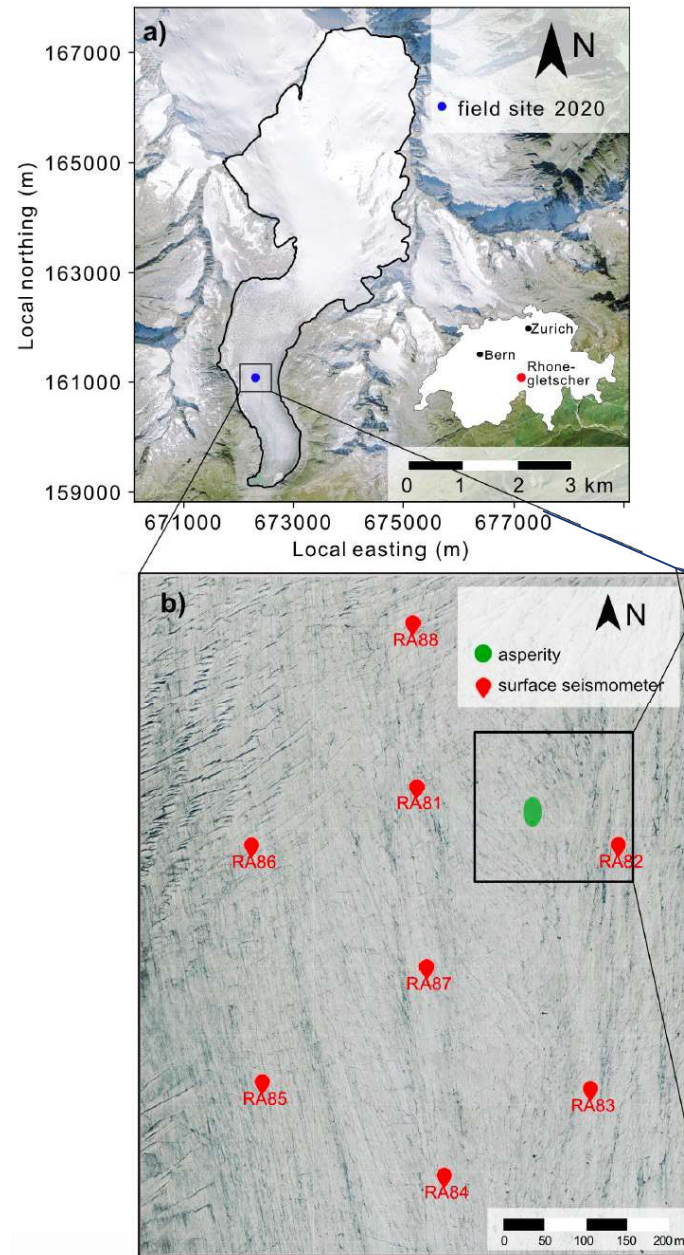
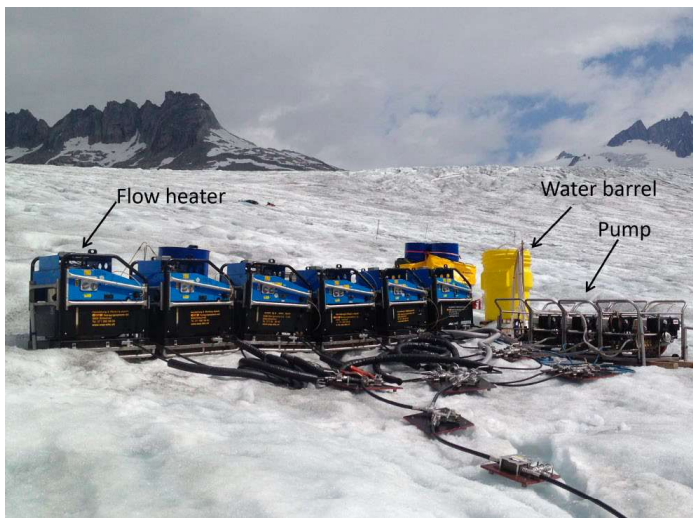
Non-Diurnal Activity Variations in Summer



- High moment rate release during low sliding velocities
- Stick-slip asperities work together.
- Different interplay between effective pressure:
Seasonal vs. sub-seasonal variations

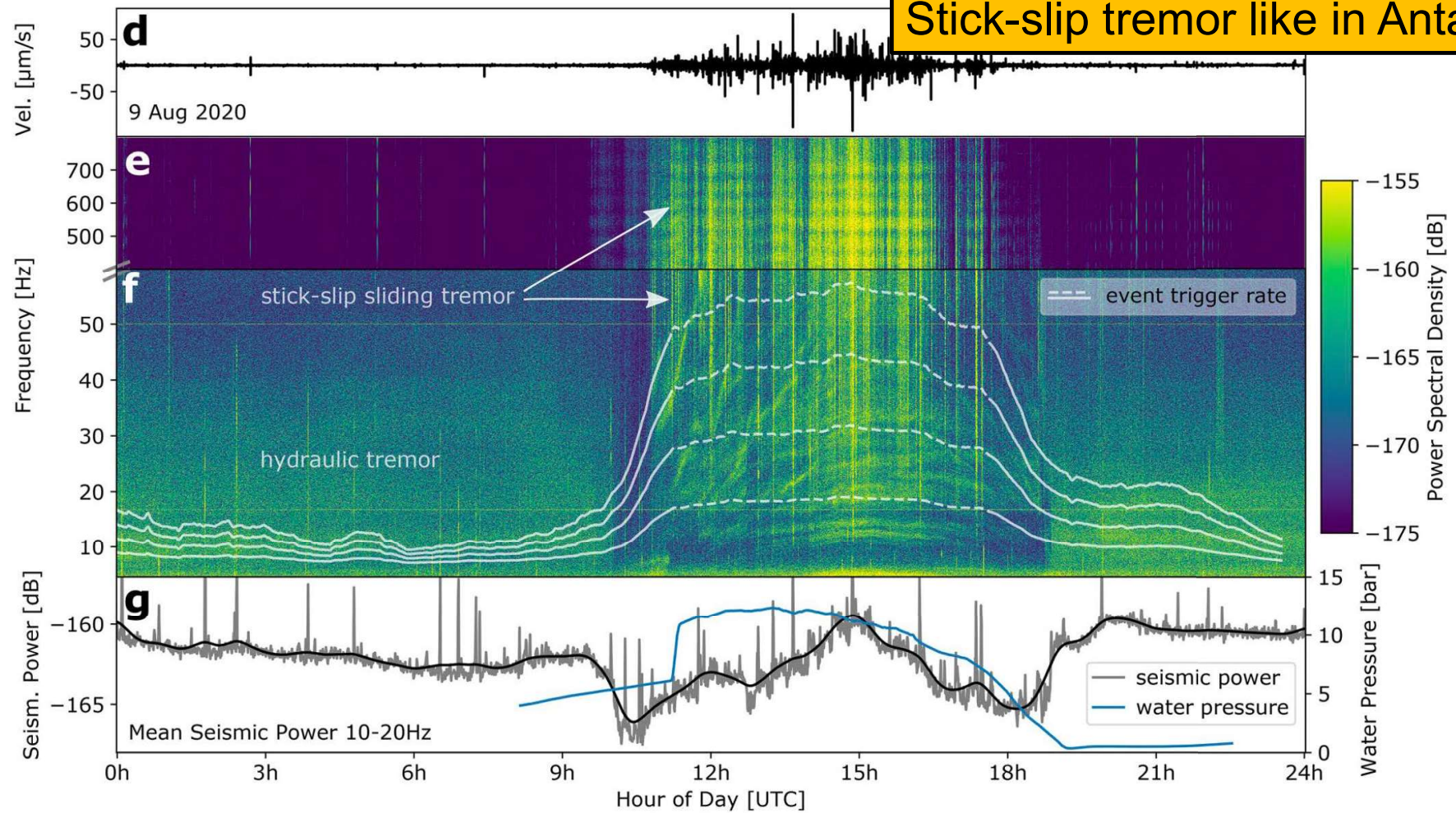
So what? Need stress drop, slip distance, rupture sizes, ...
See work by Tom Hudson

Study Site

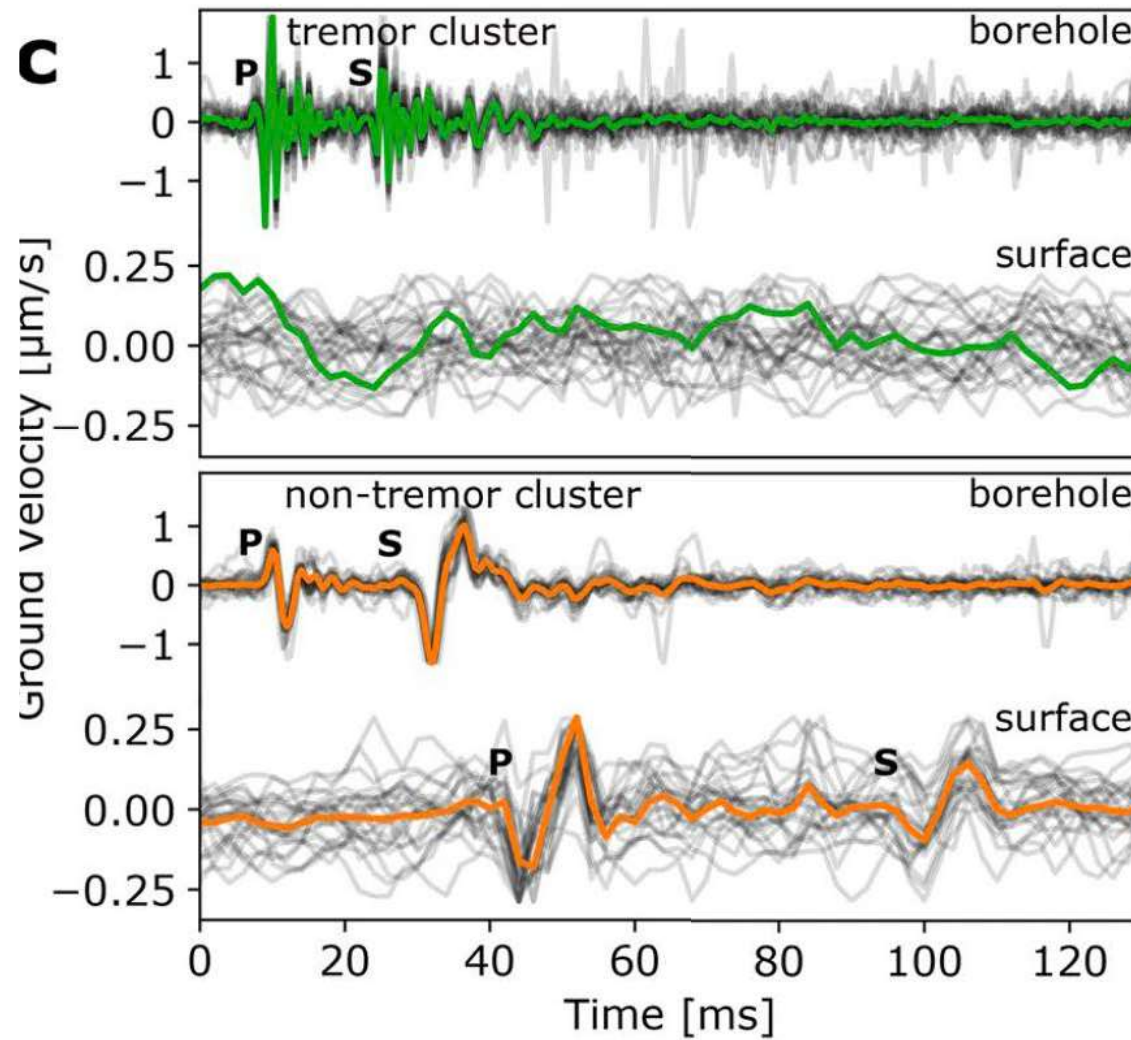


Gliding Spectral Lines

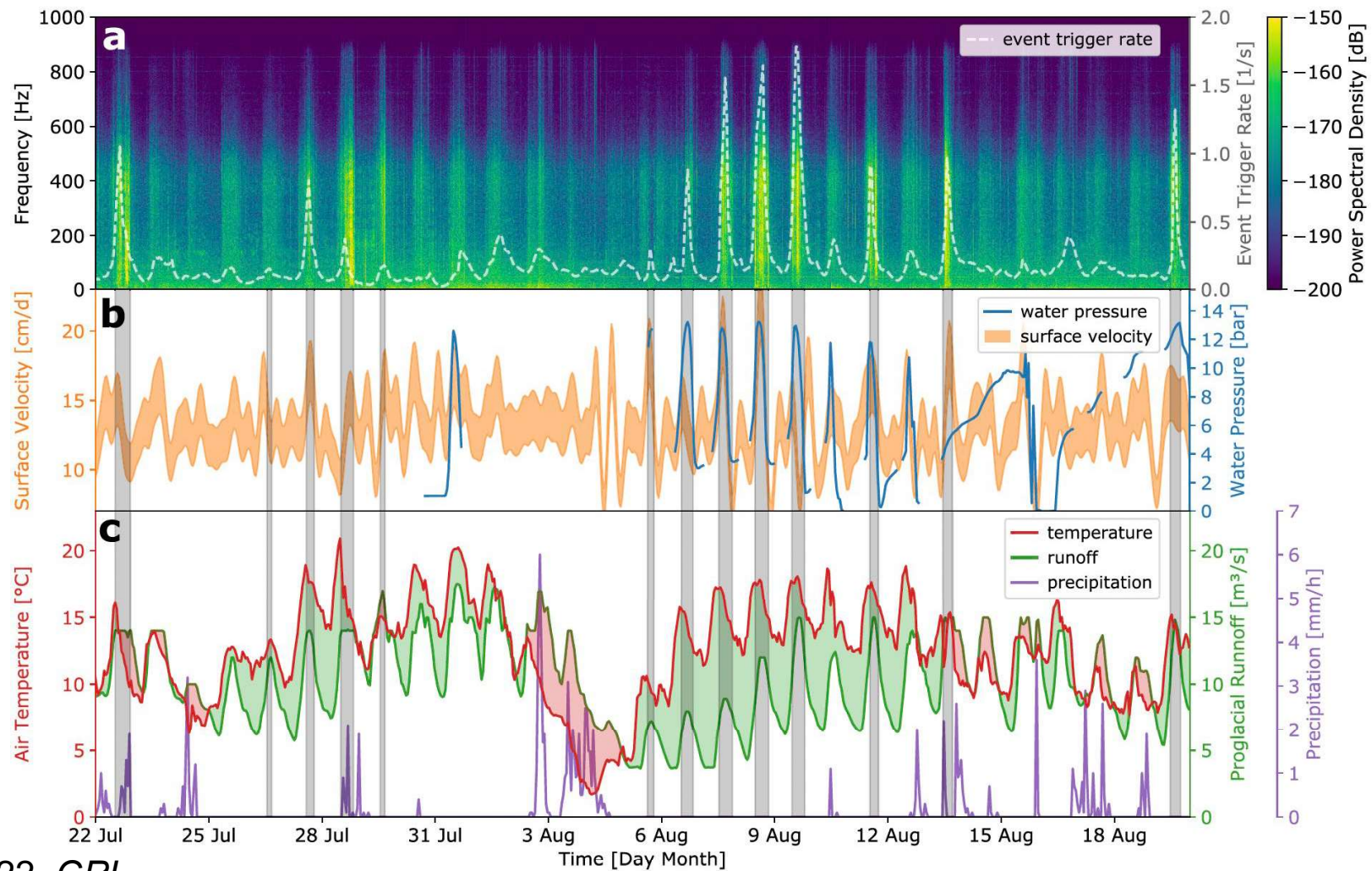
Stick-slip tremor like in Antarctica.



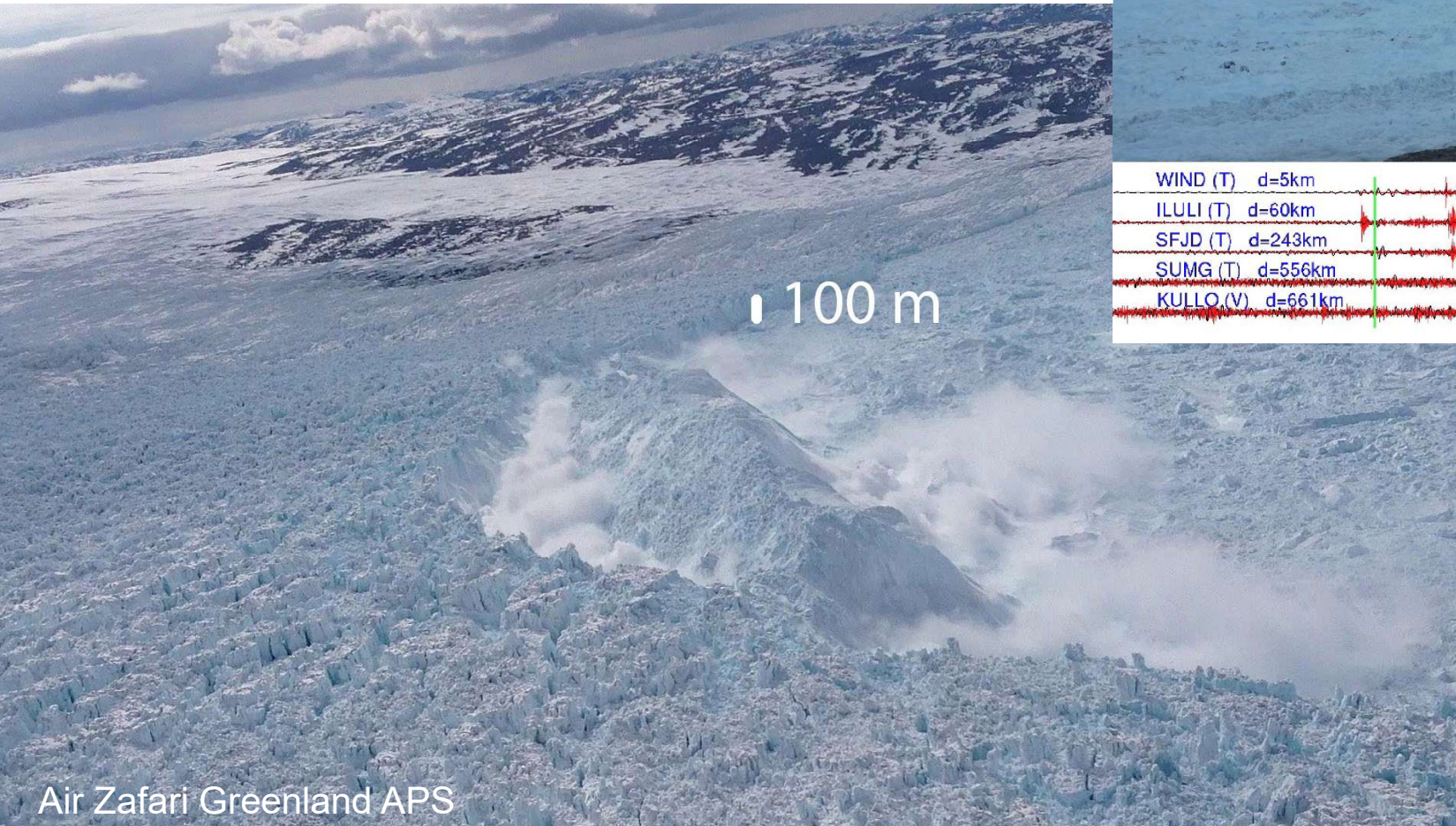
Tremor vs. non-Tremor Events



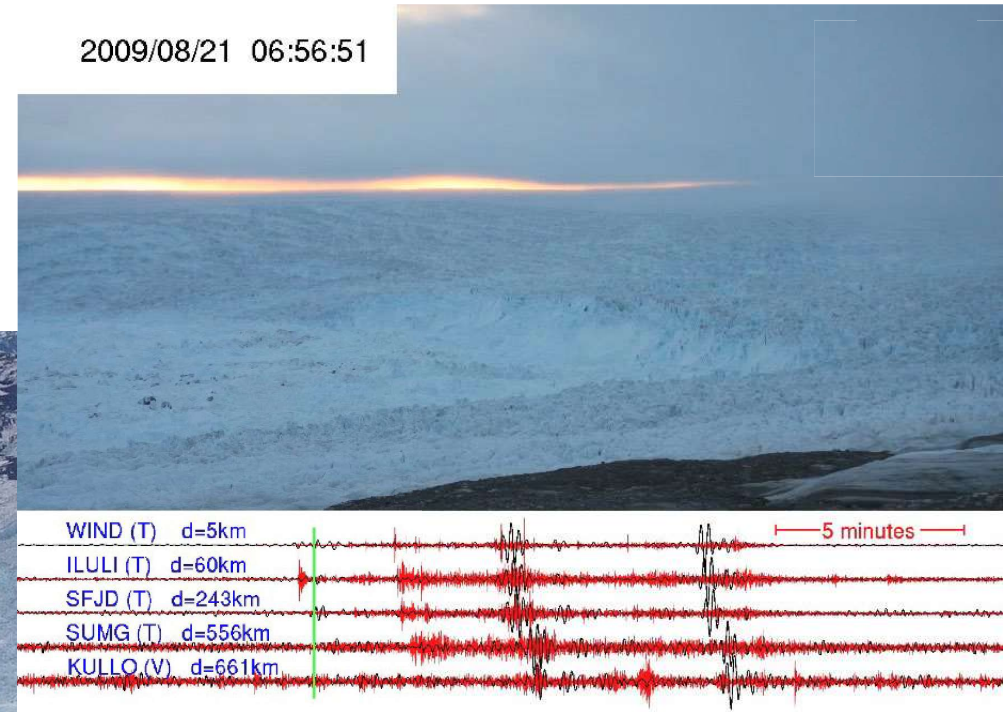
Stick-Slip Tremor



Iceberg Calving

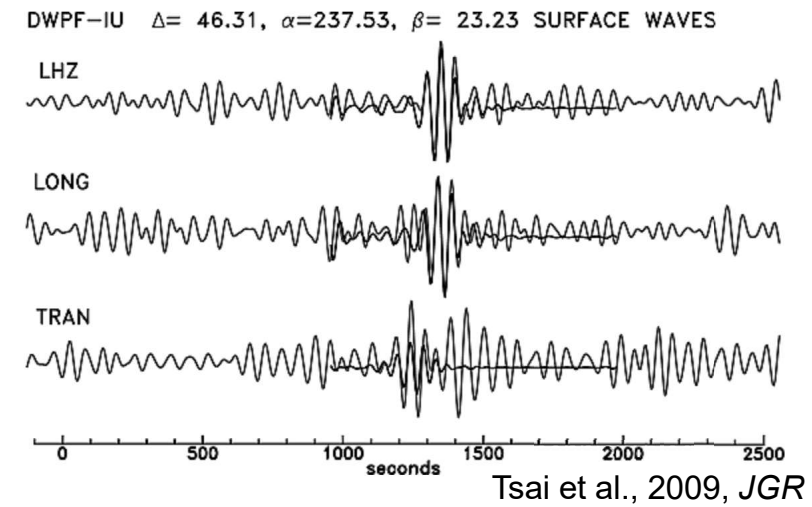
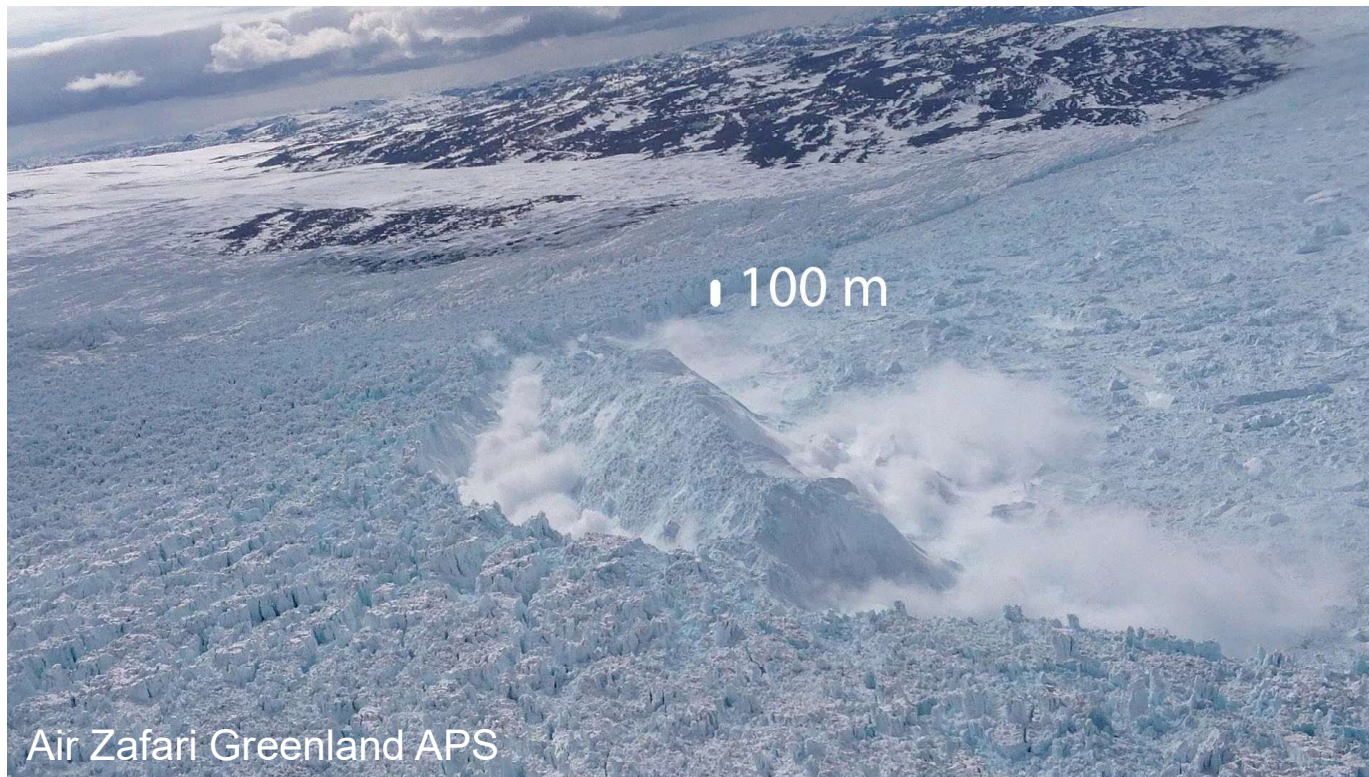


2009/08/21 06:56:51

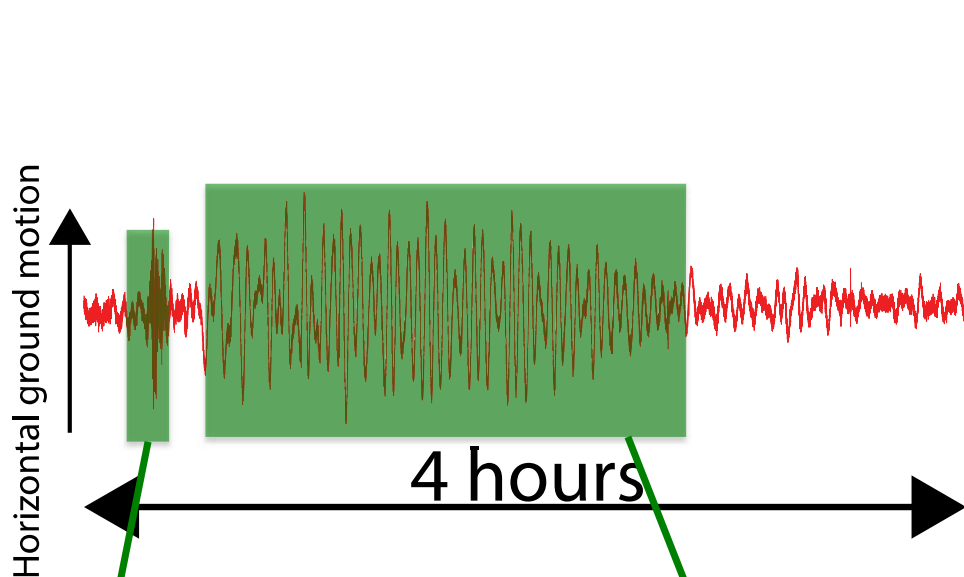


Walter et al. 2012, JGR

Glacial Earthquakes

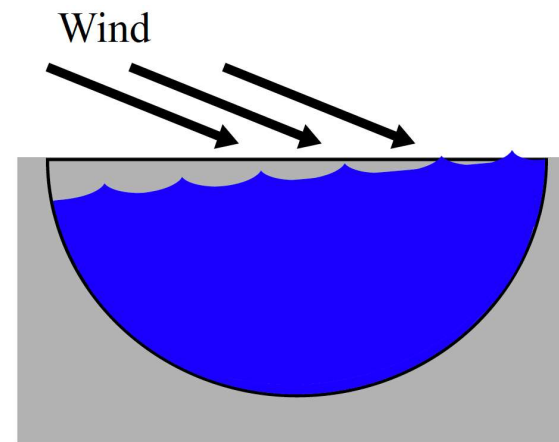


Broadband Calving Seismicity

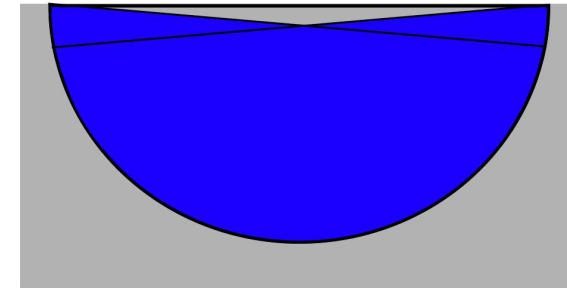


Fjord seiche

- Standing water wave in fjord
- 100-1000 s
- Lasting for hours



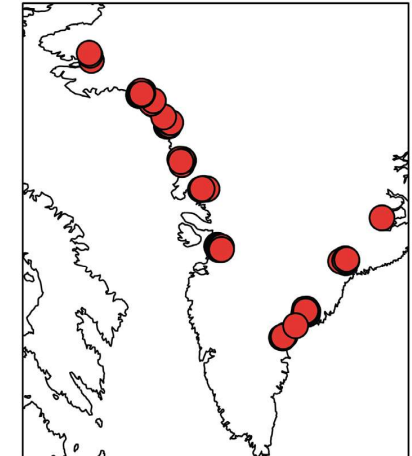
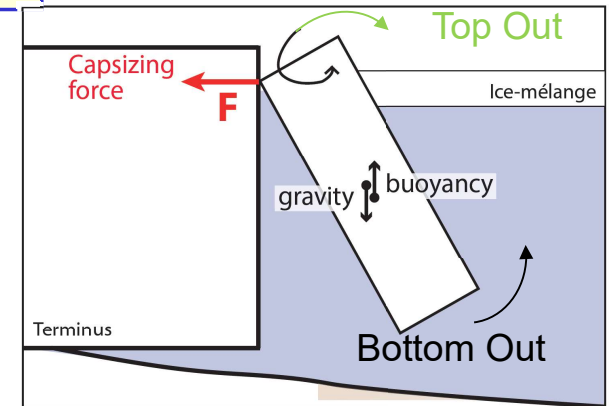
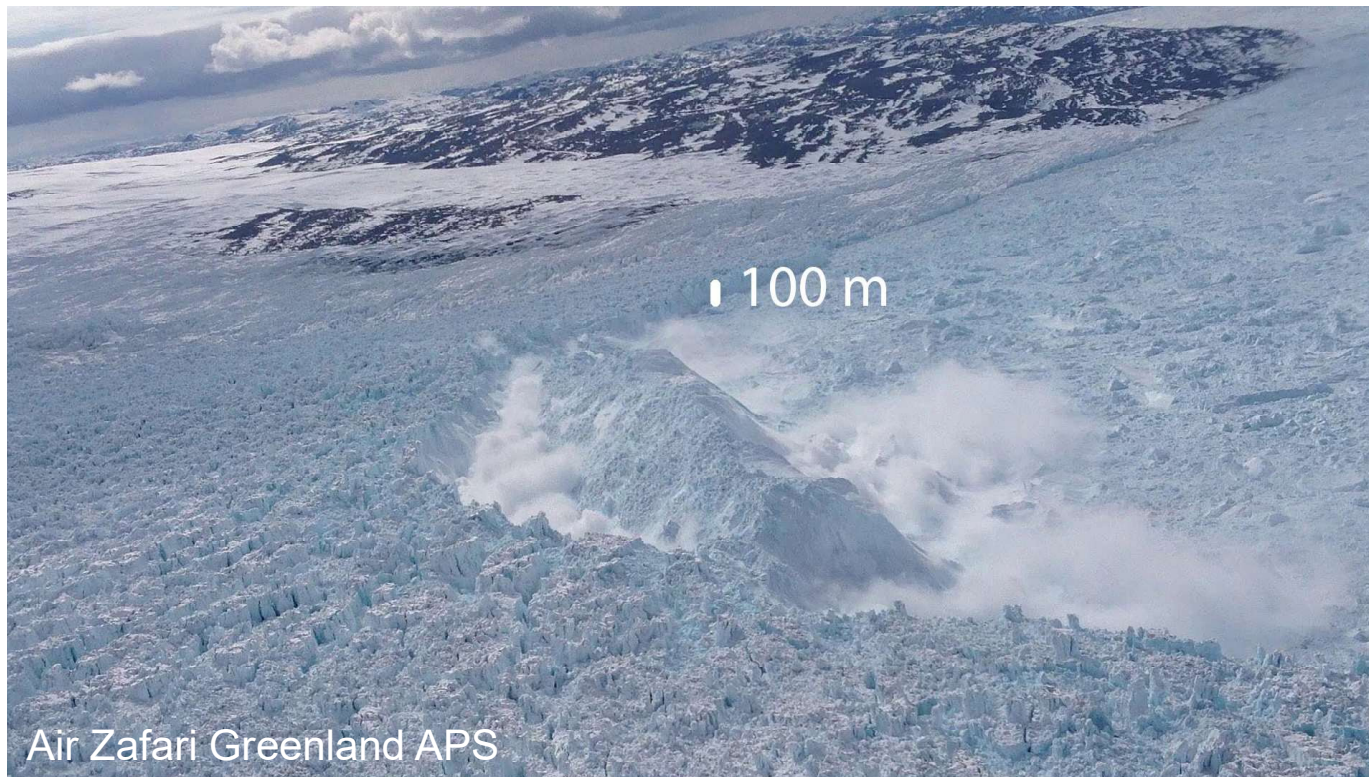
Oscillations of lake surface



Von Smokeonthewater in der Wikipedia auf Deutsch - Eigenes Werk (Originaltext: eigene Arbeit).
CC BY-SA 3.0 de, <https://commons.wikimedia.org/w/index.php?curid=20452351>

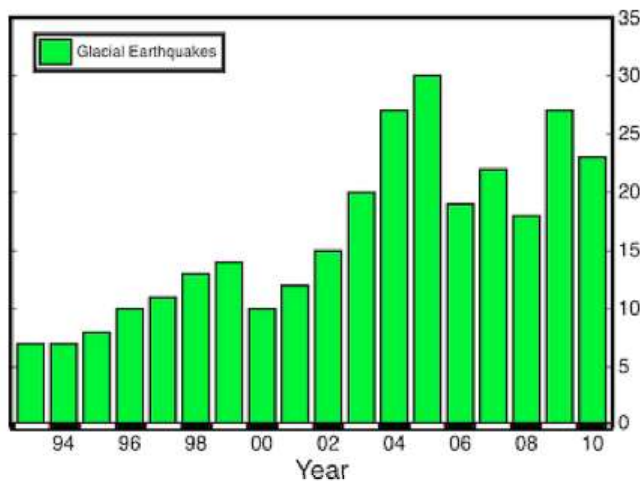
Seismic signal generated by iceberg calving

See work by Anne Mangeney



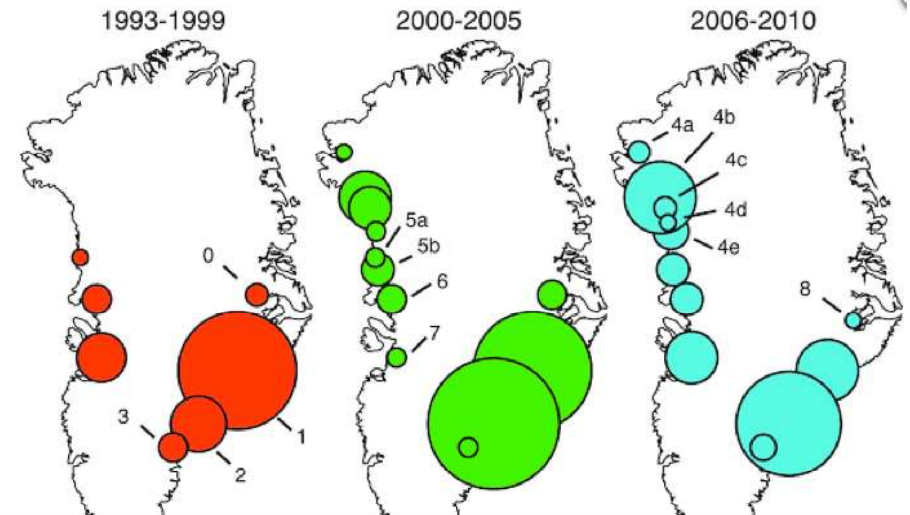
Veitch and Nettles 2012, Sergeant et al., 2016

Glacial Earthquakes (Ekström et al., 2003)



Temporal Activity of Glacial Earthquakes

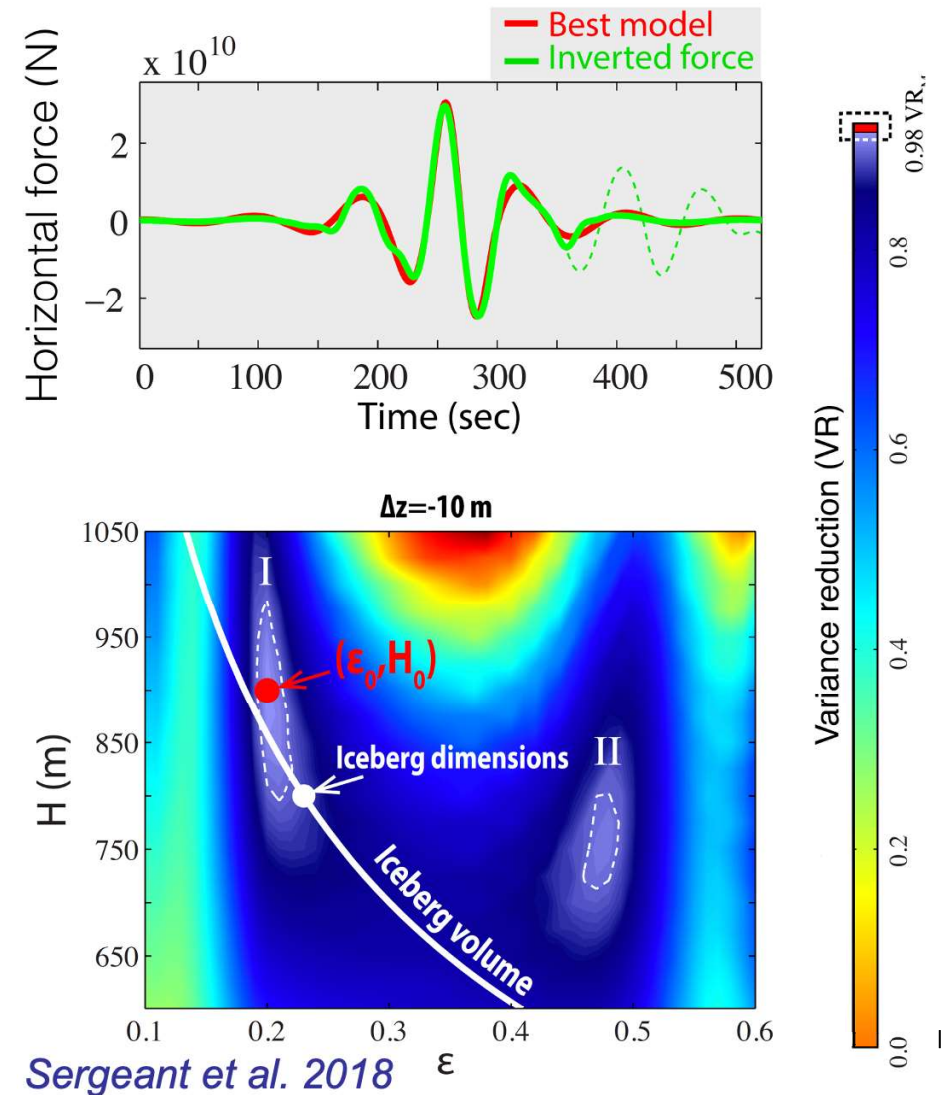
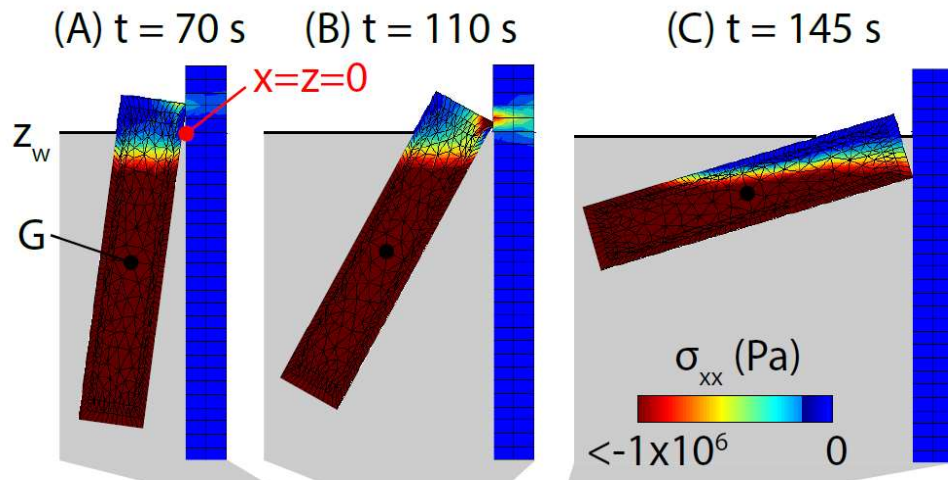
- Increase since the 90's
- Seasonality: peak during summer months



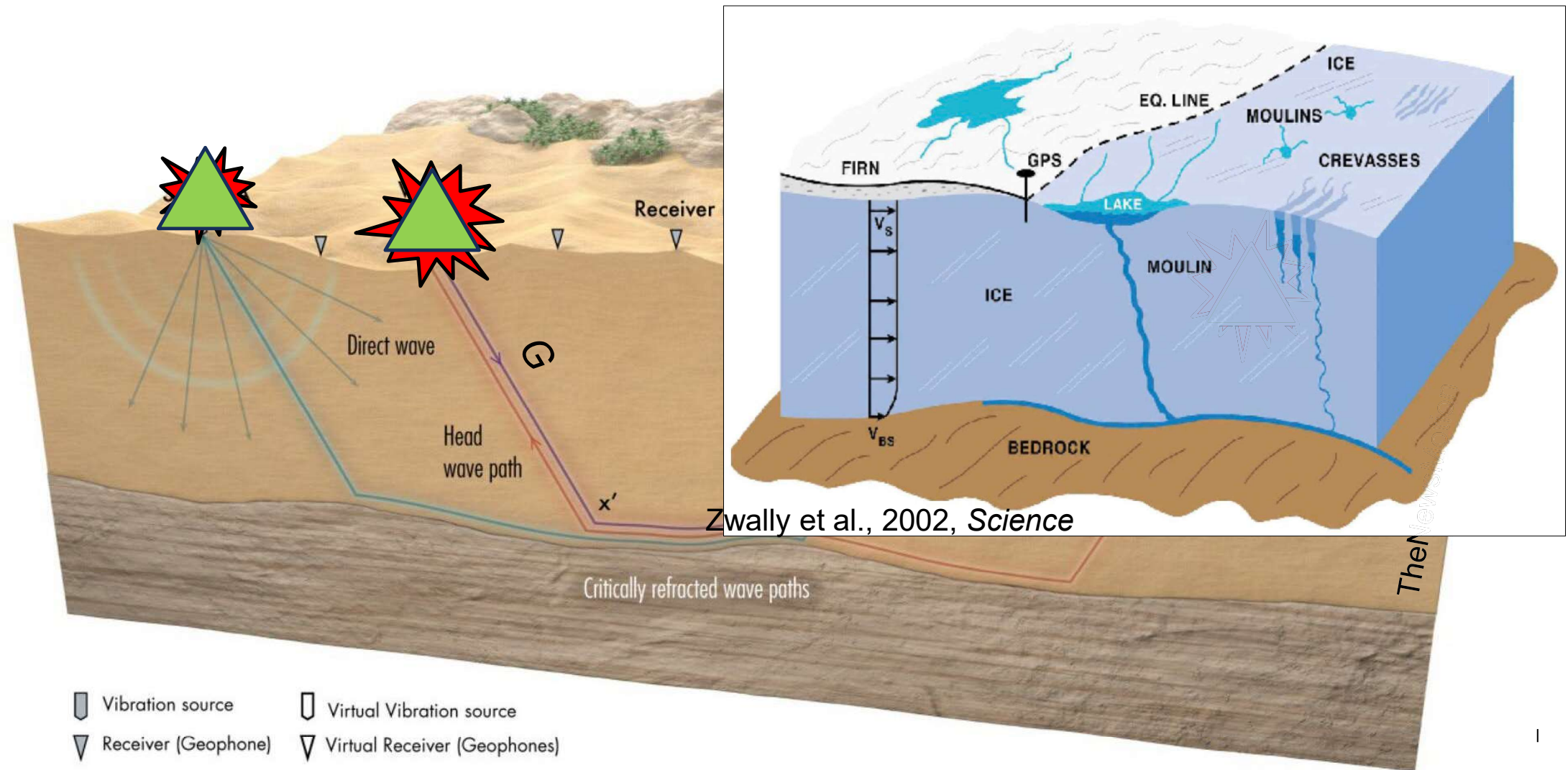
Geographic Spreading of Glacial Earthquakes

- Northward shift
- Corresponding to warming coastal waters
- → calving activity and dynamic discharge reacts to fjord warming

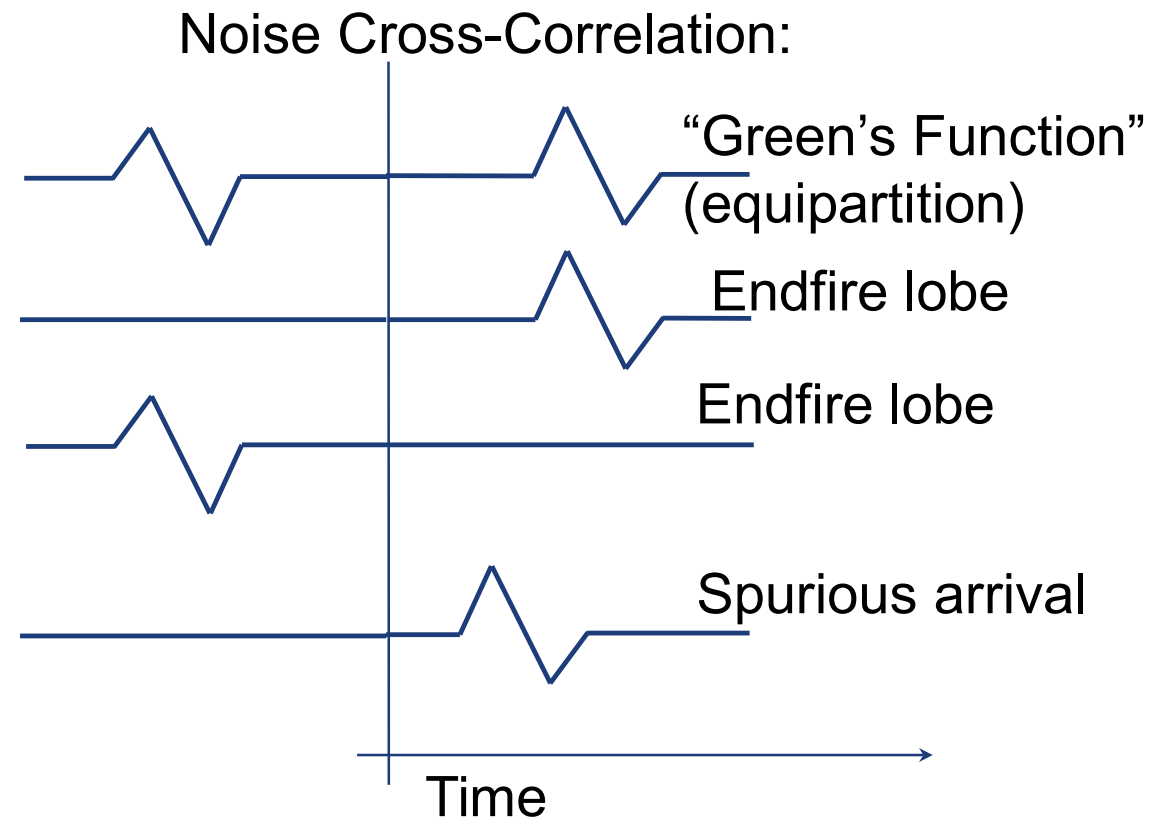
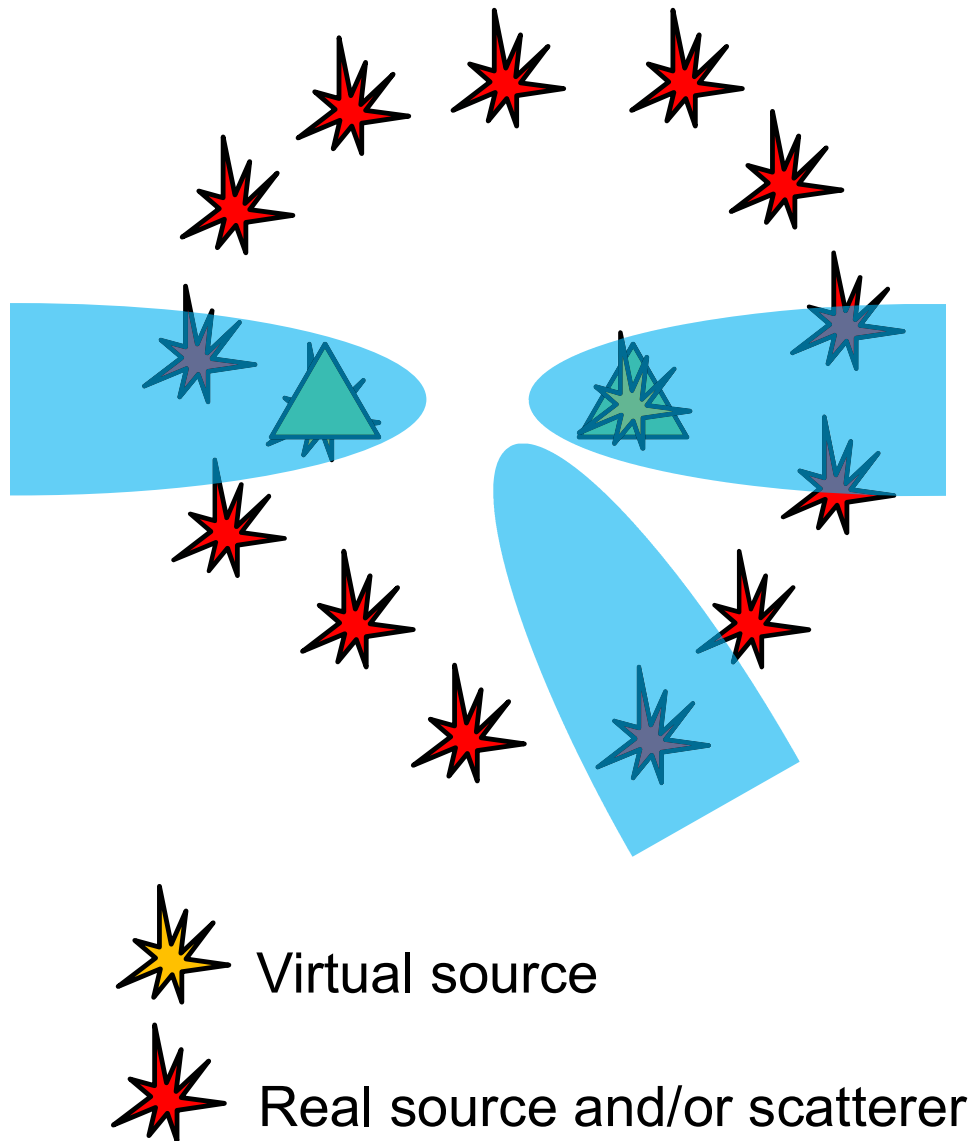
Capsize Modeling (A. Mangeney, IPGP)



Noise Correlation: Virtual Seismic Sources

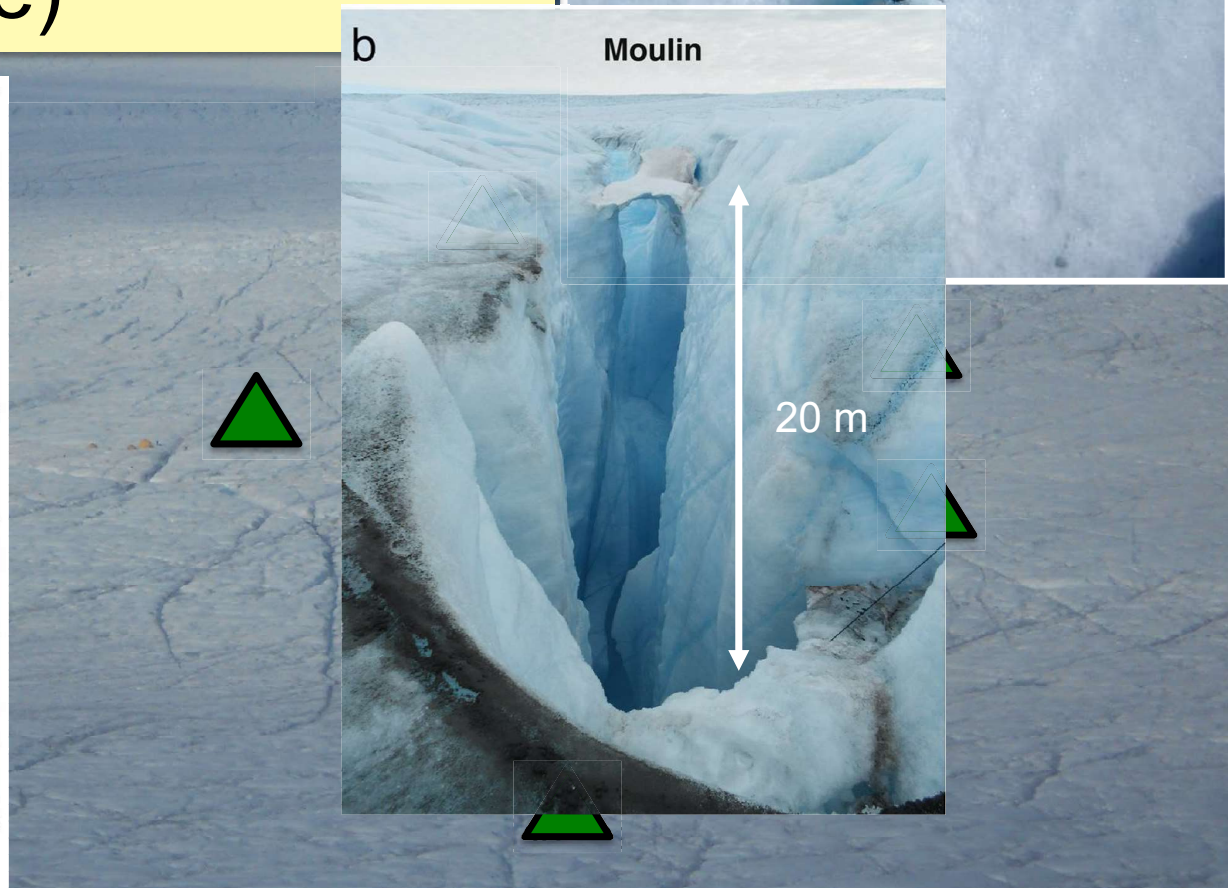
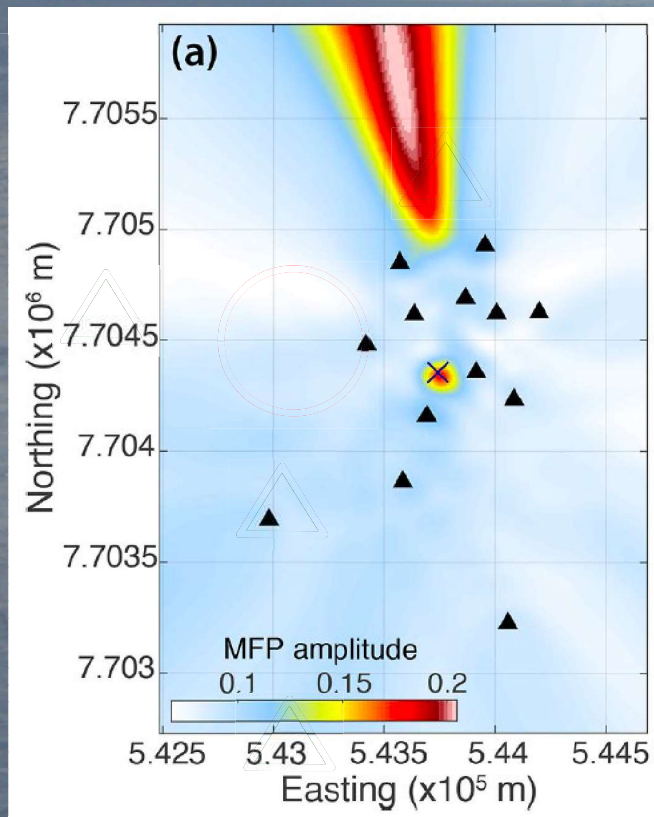


Crash Course: Seismic Noise Interferometry



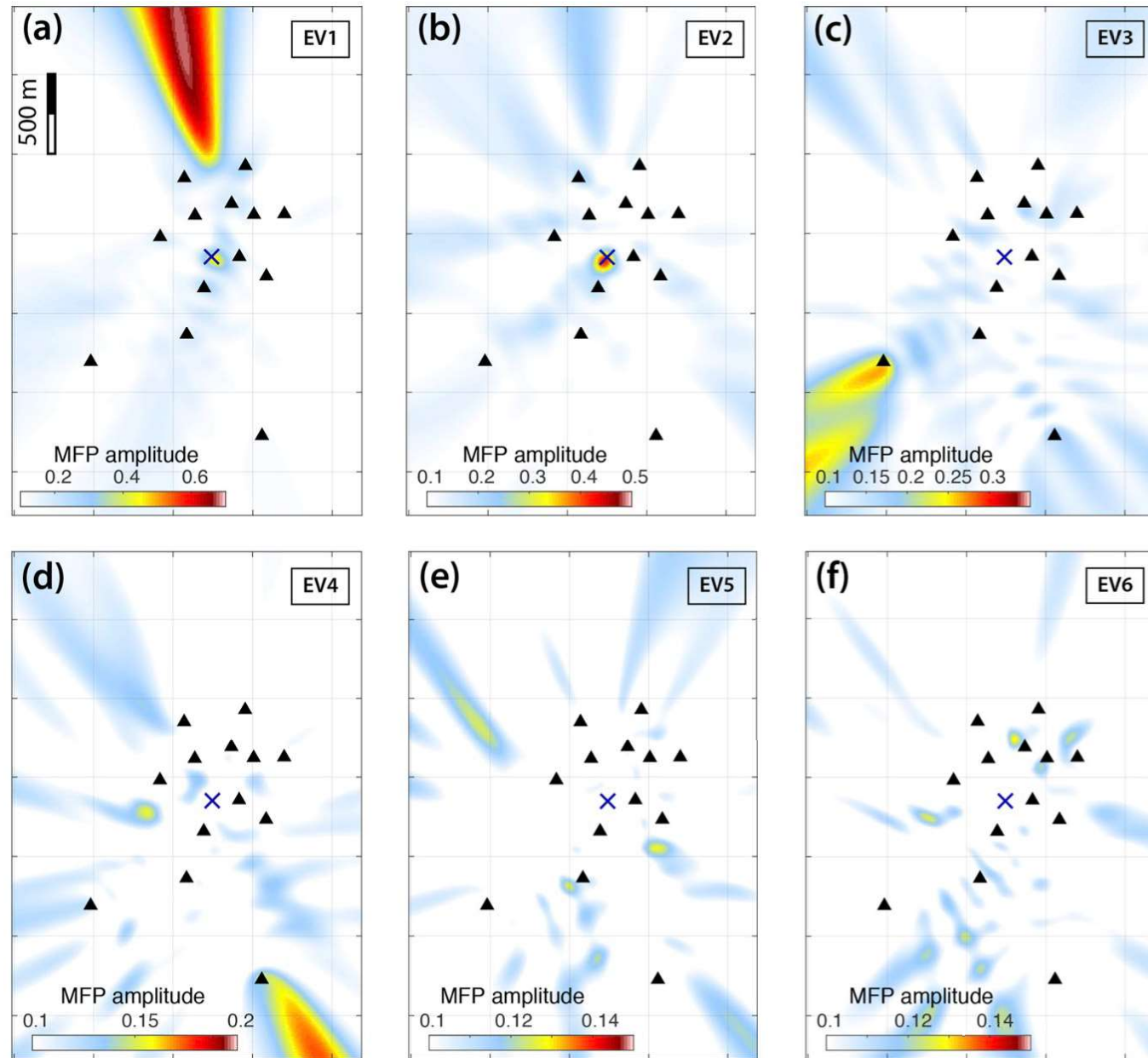
Seismic Monitoring on the Greenland Ice Sheet

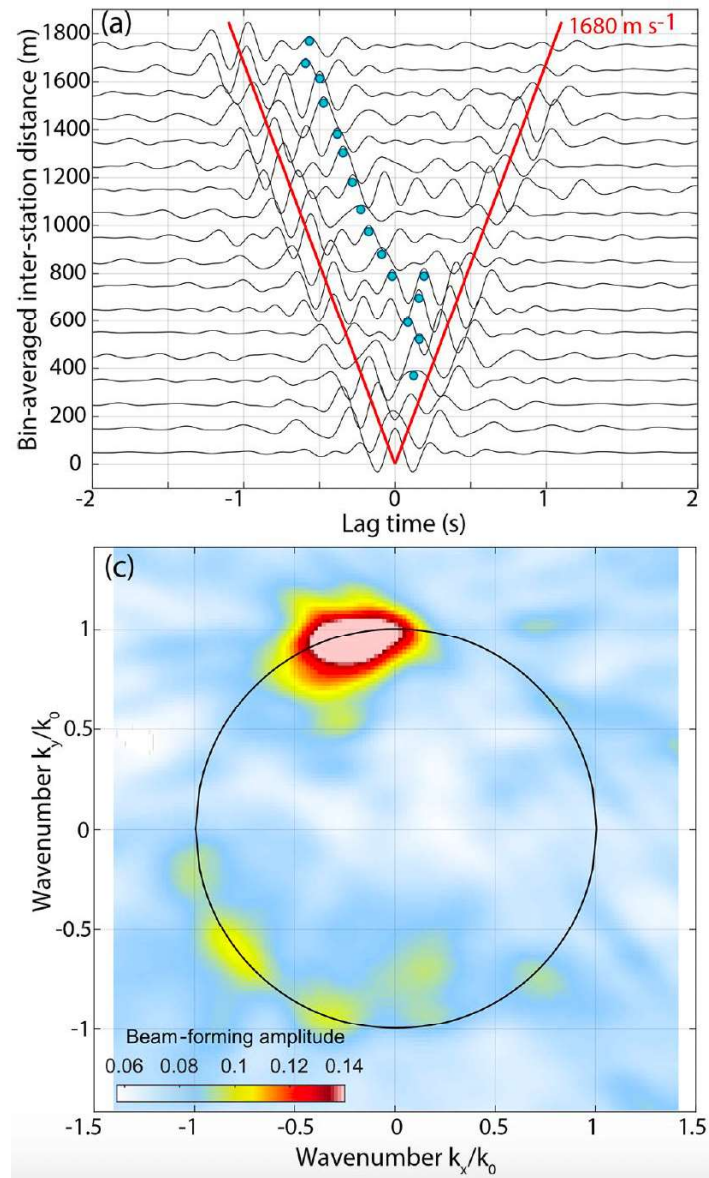
Beamforming:
Use network coherence
(antenna principle)



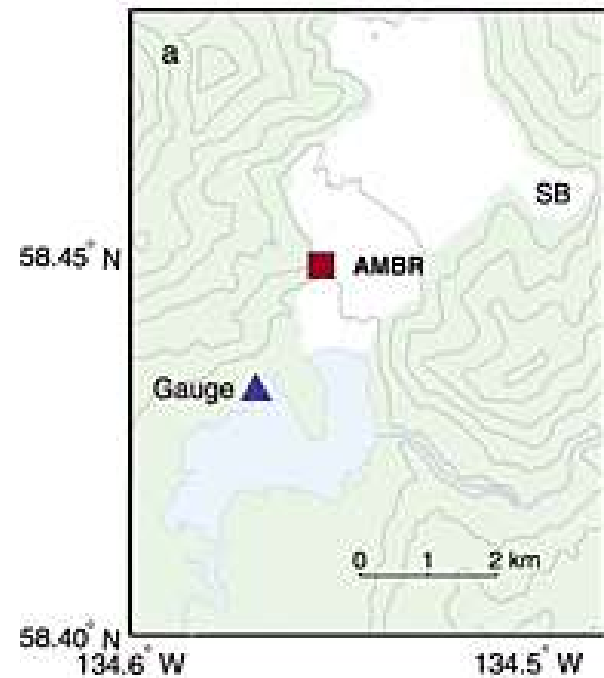
Roesli et al., 2016, JGR
Sergeant et al., 2020, *The Cryosphere*

Noise source separation with singular value decomposition

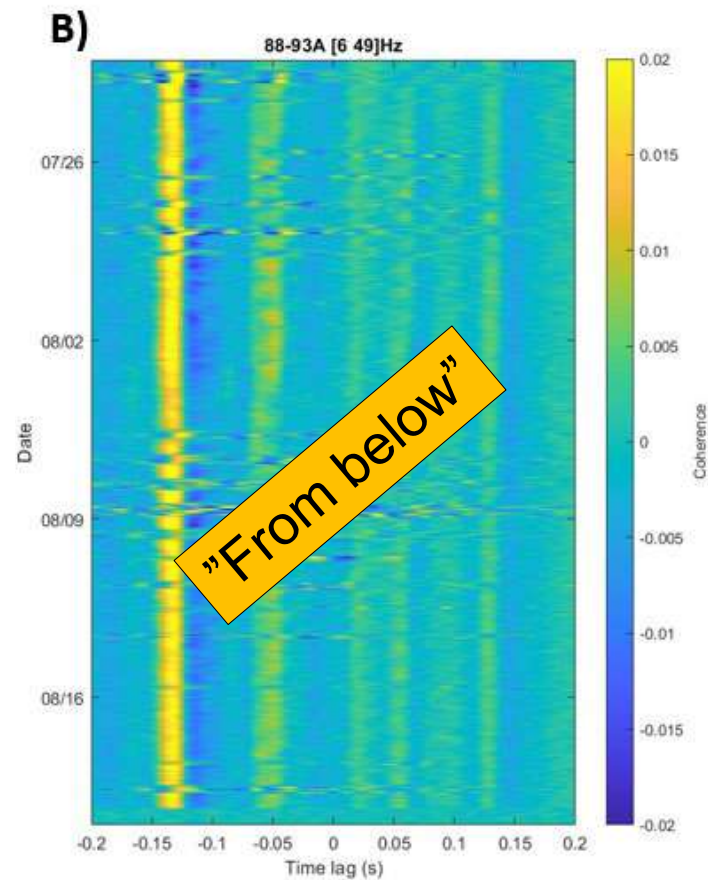
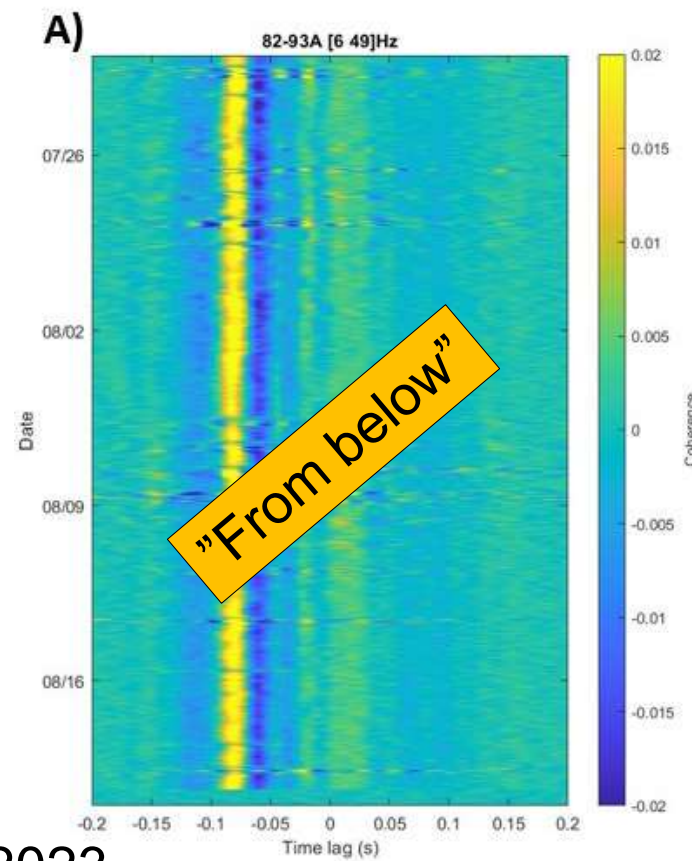
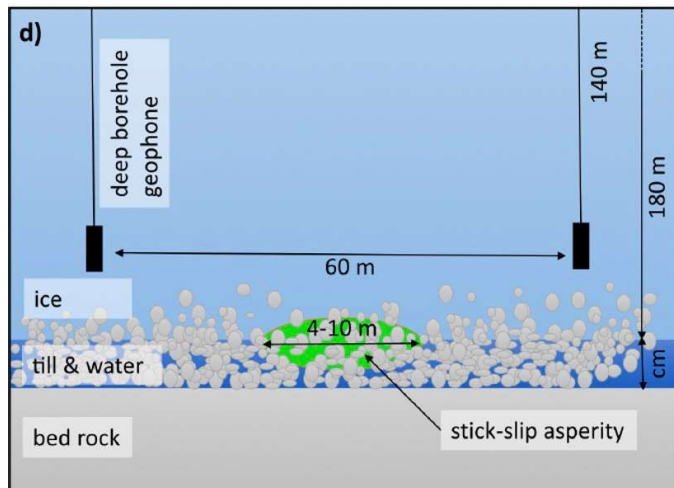




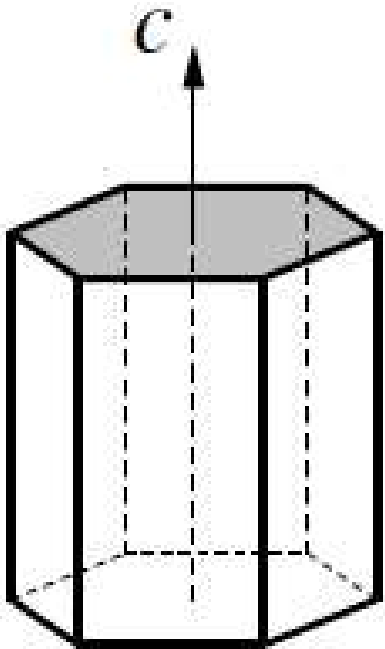
Subglacial discharge



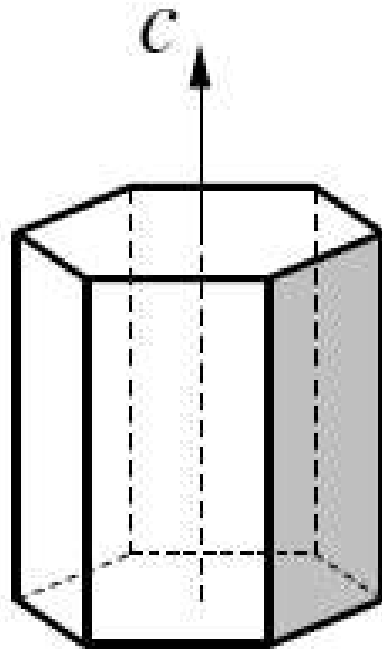
Surface-Borehole Interferometry



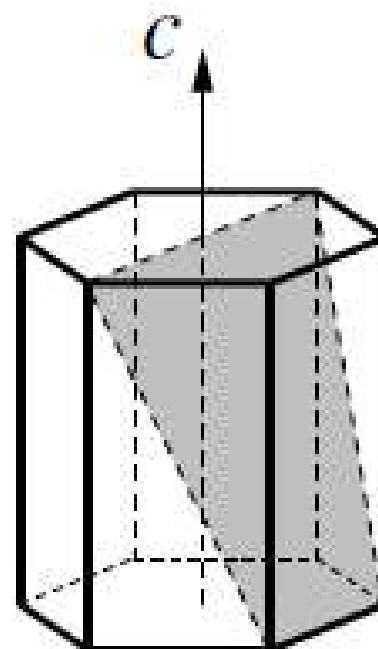
Ice Crystal Structure and Elastic Wave propagation



Basal

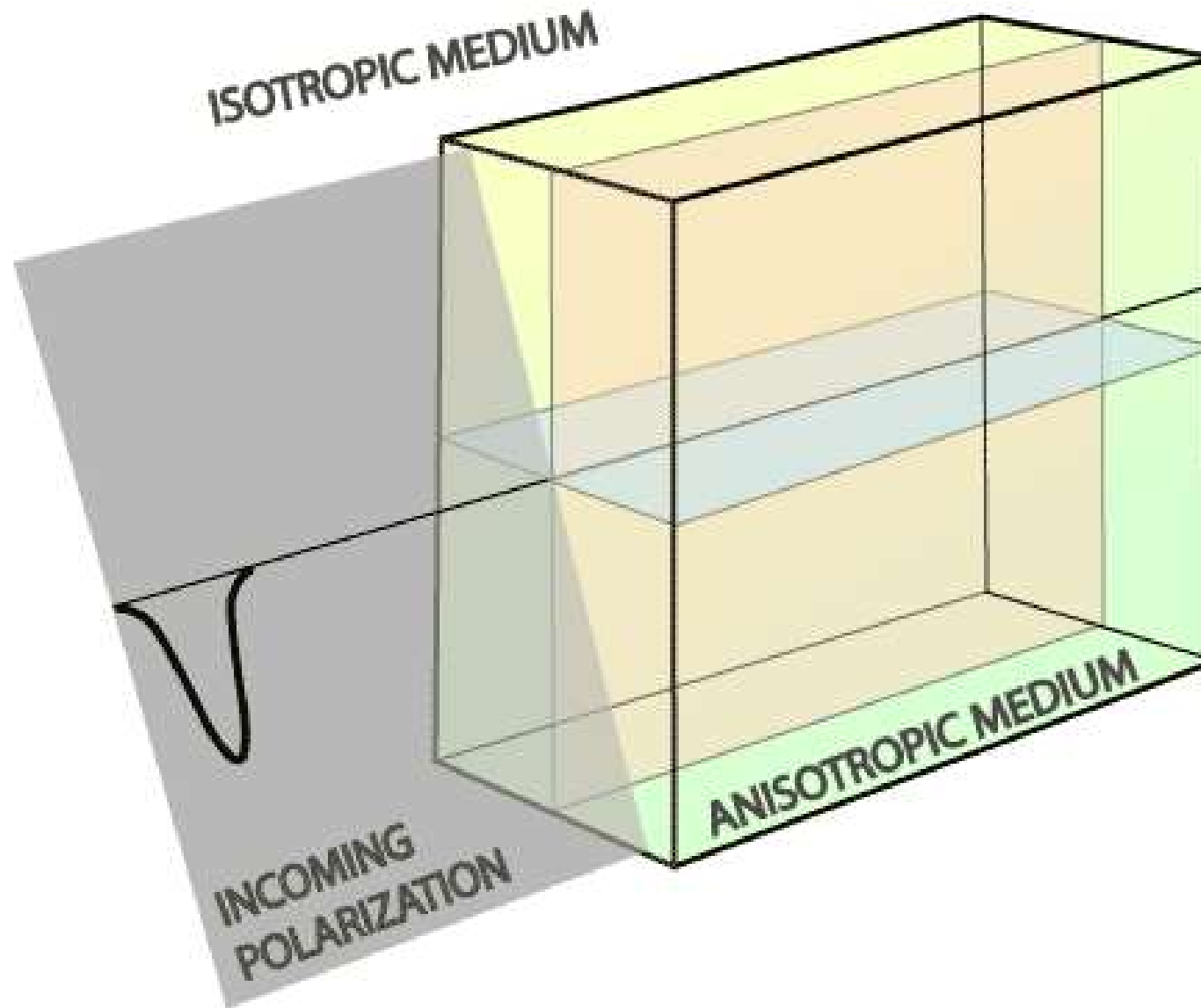


Prismatic

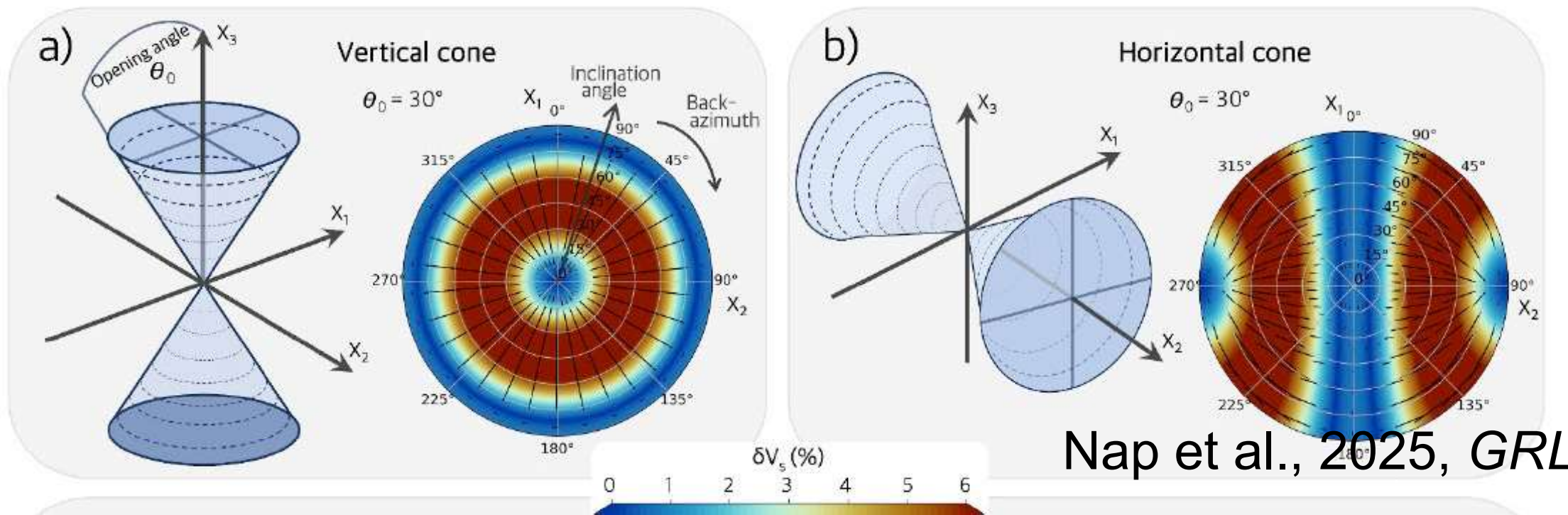


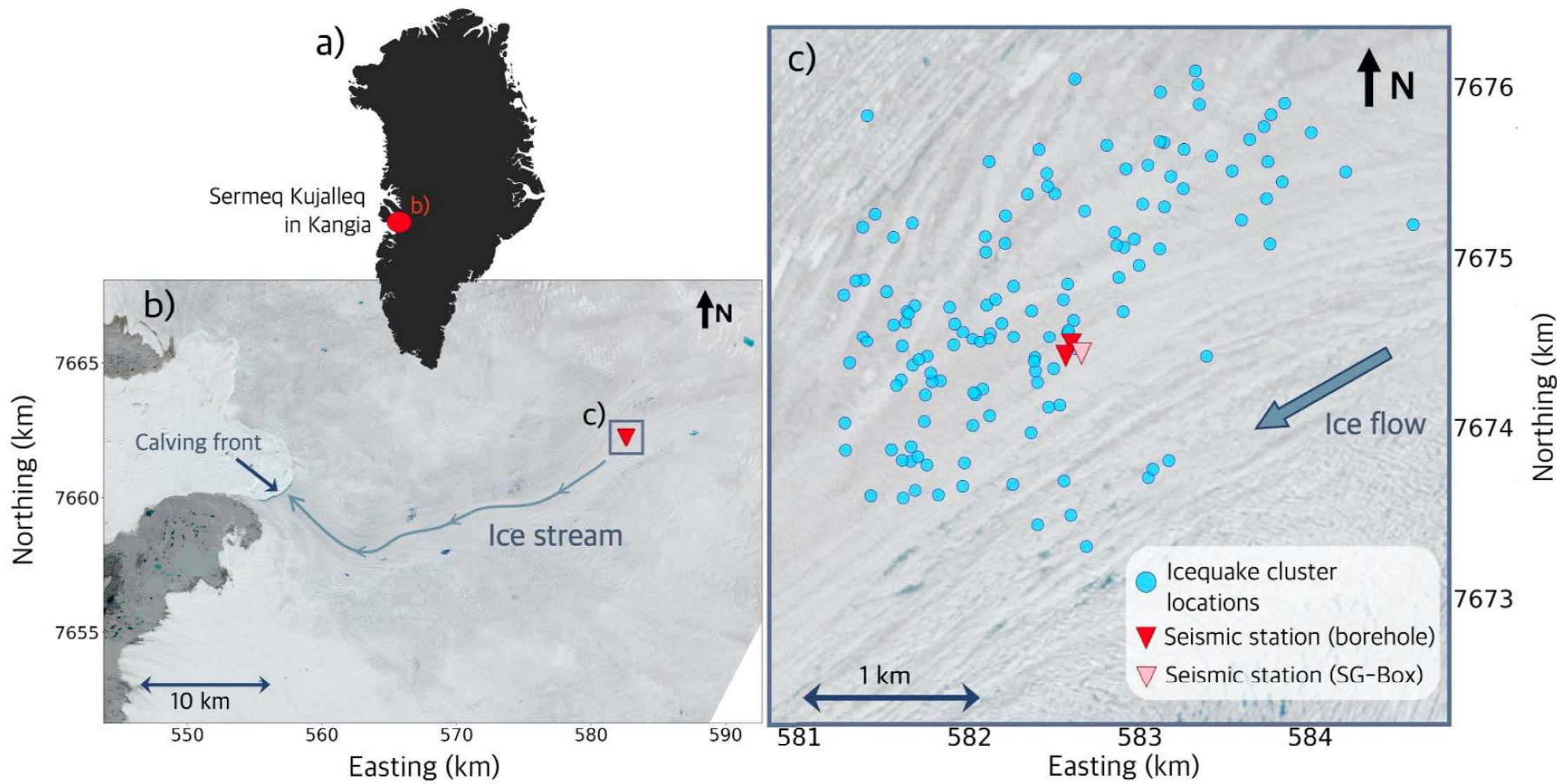
Pyramidal

Shear Wave Splitting

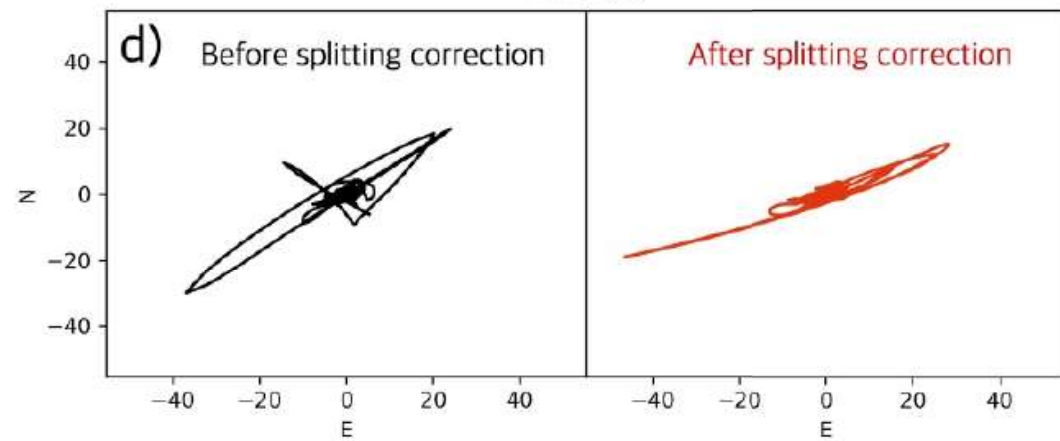
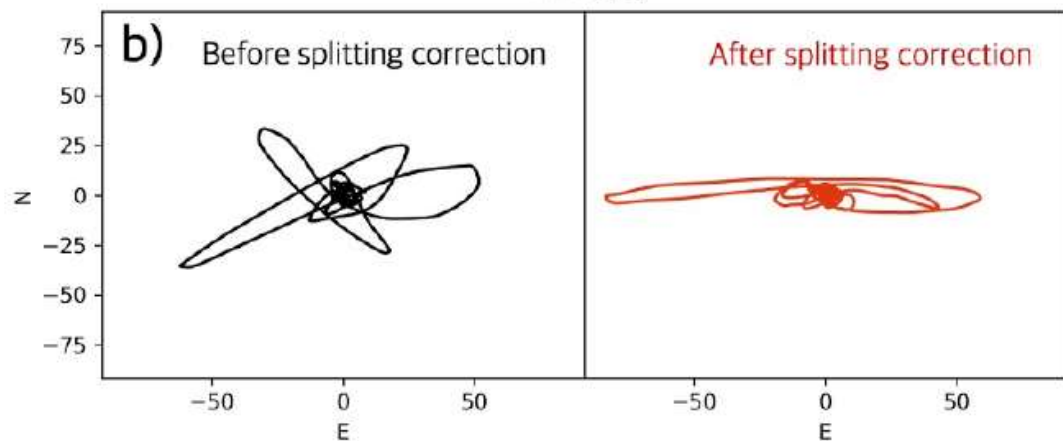
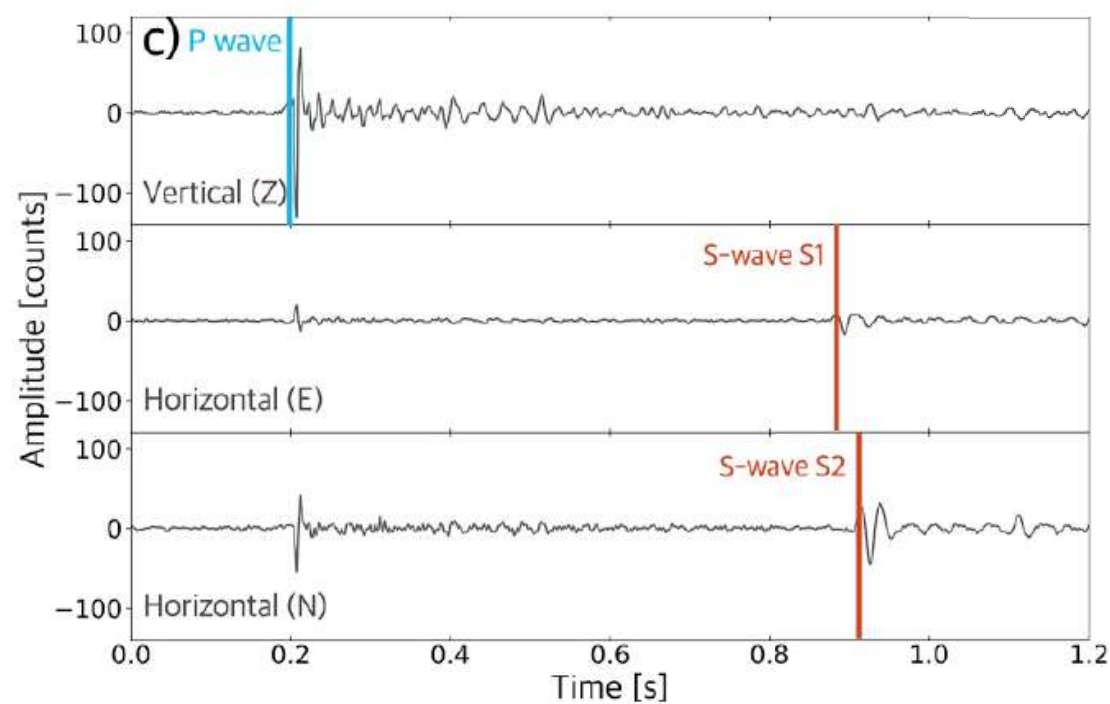
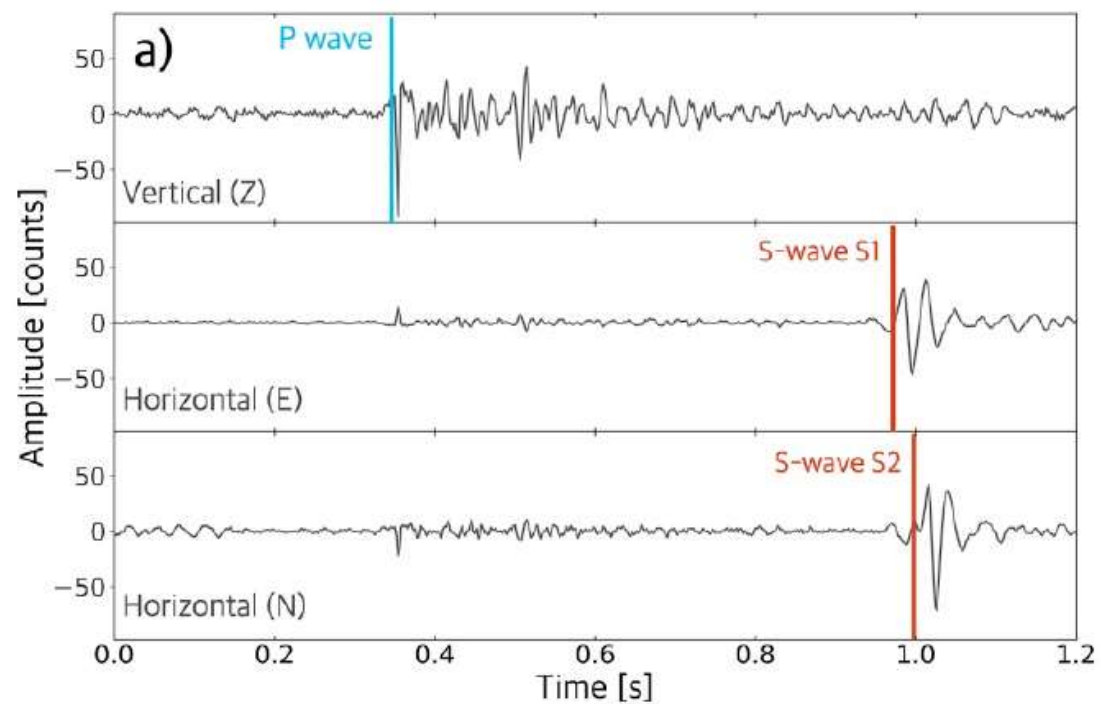


Shear Wave Splitting and Fabric

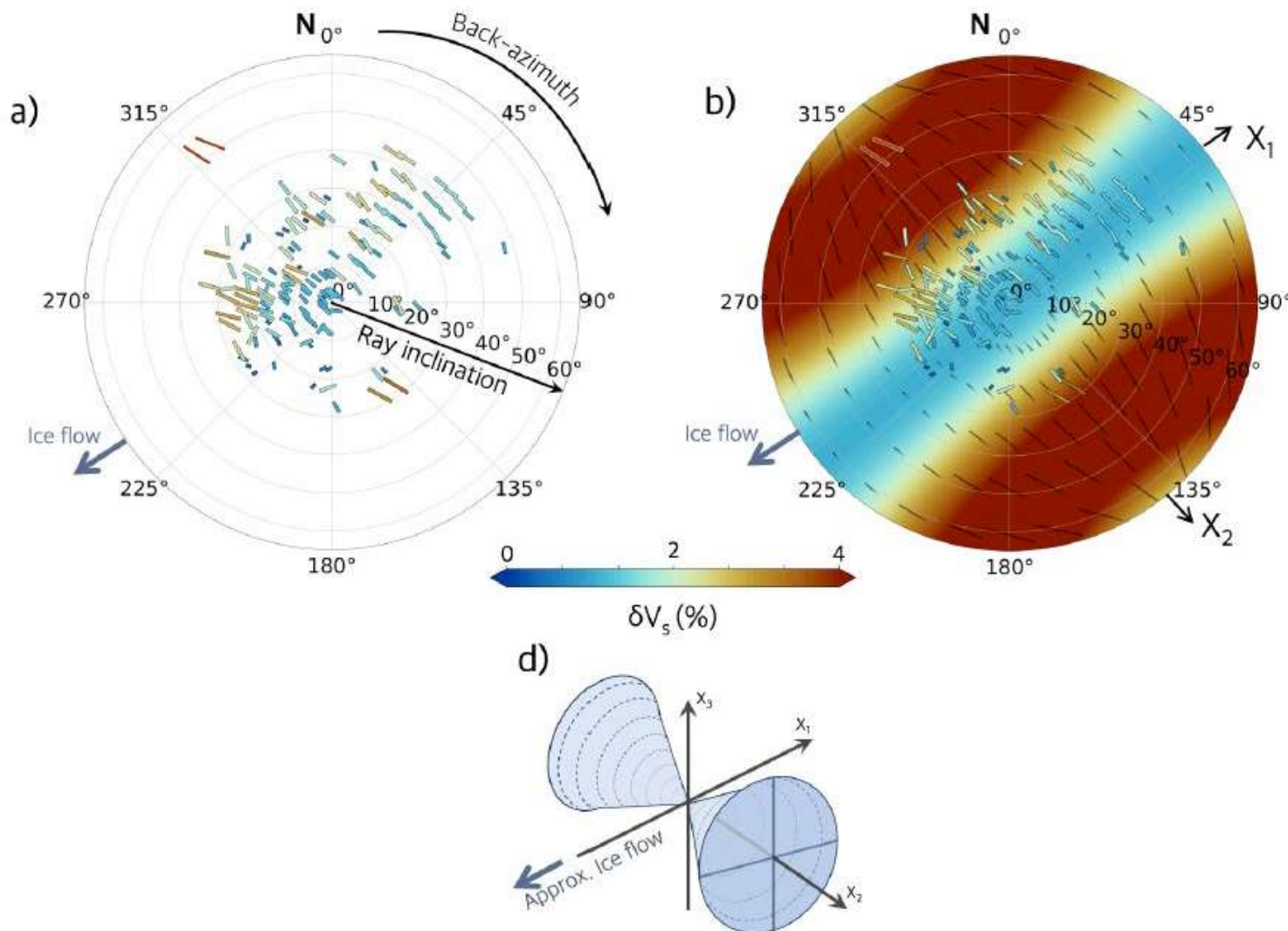




Nap et al., 2025, *GRL*



Propagation through 2-2.5 km thick ice



Simple shear in X_1 - X_3 plane

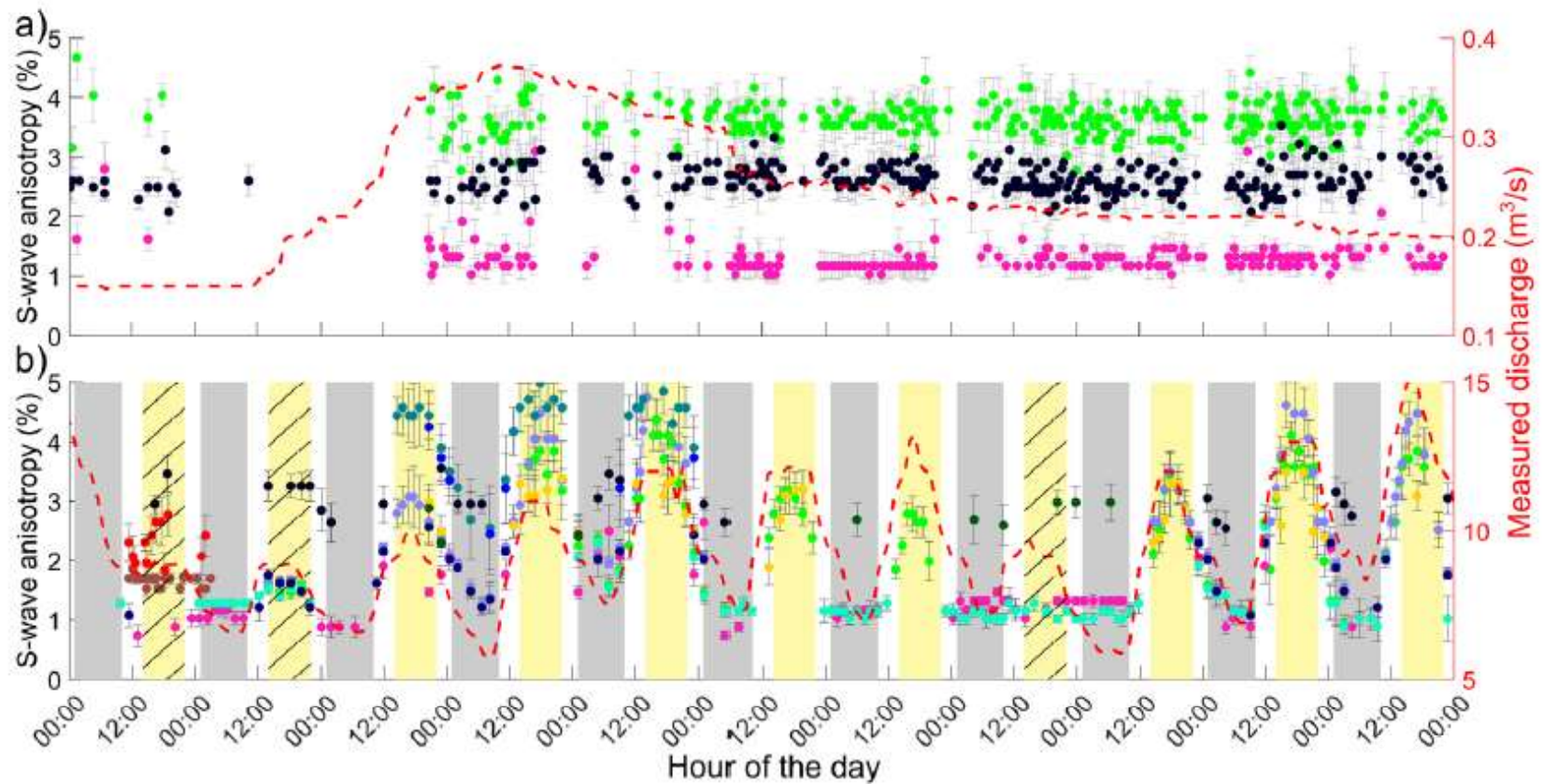
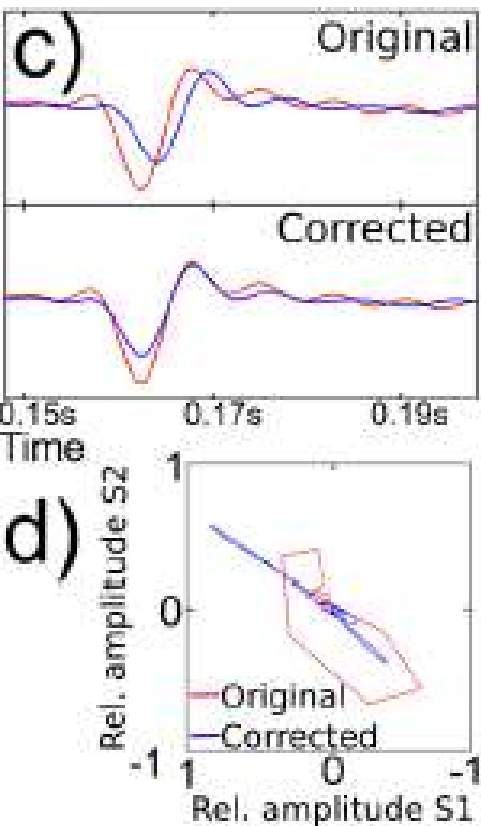
OR

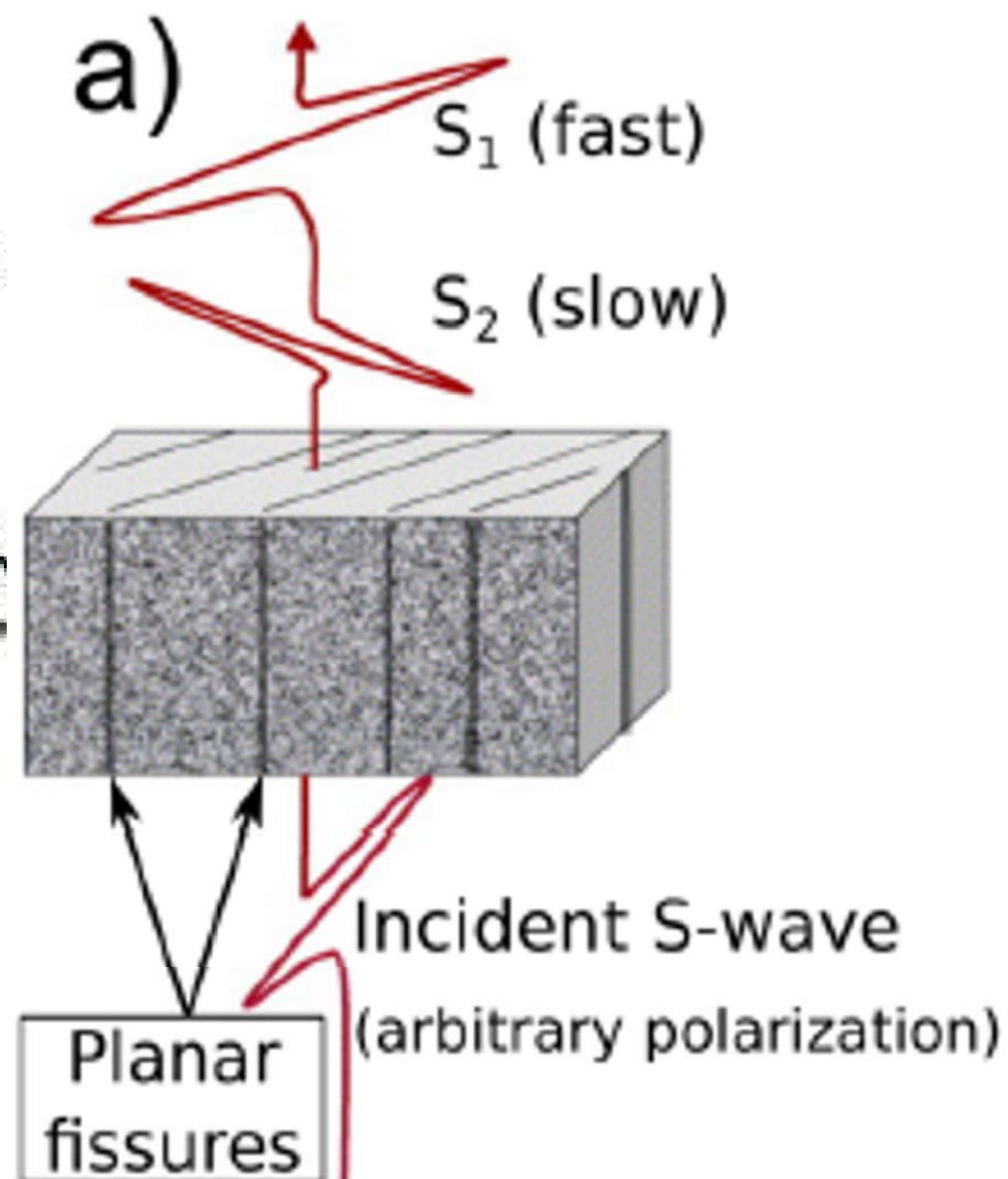
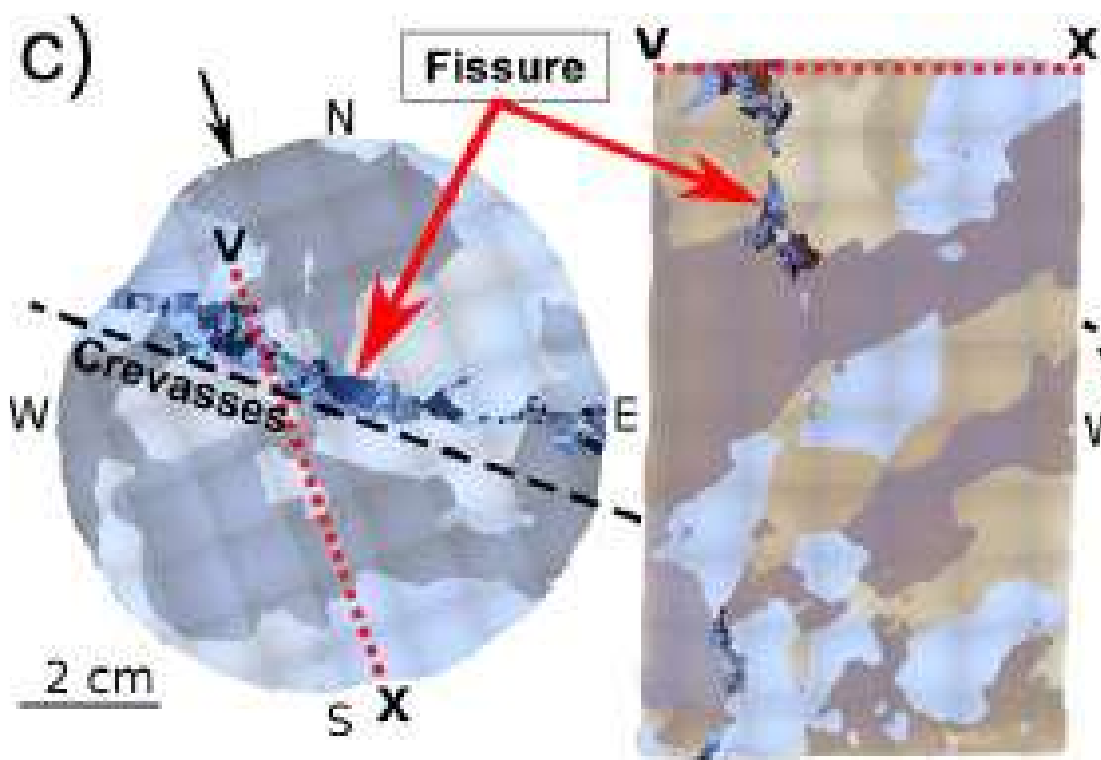
Uniaxial compression along X_2

Deformation along-flow is at least **seven** times easier than across-flow.



Time-Varying Anisotropy on Rhônegletscher (Gajek et al., 2021)

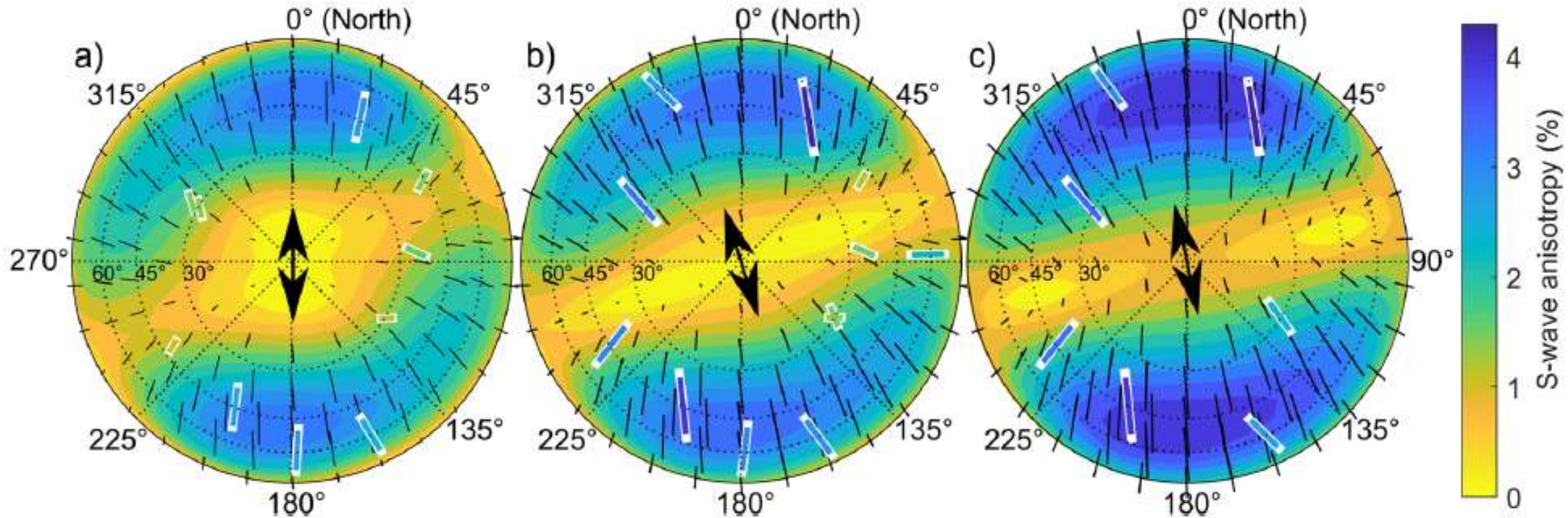




Low SWS

High SWS

Highest SWS

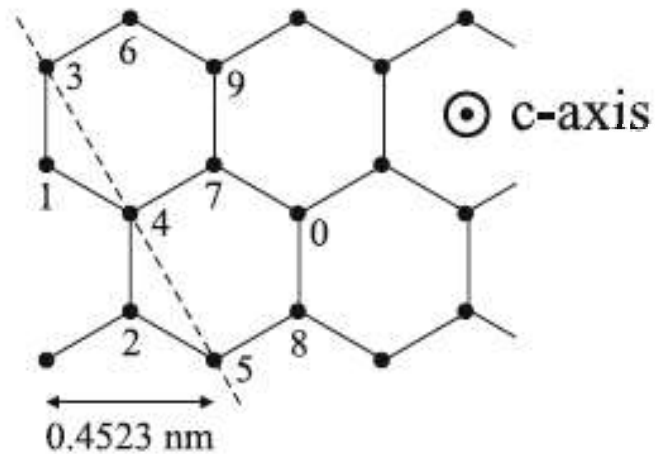


- 0.9% crack density during low-pressure times
- 90–180% increase in storage volume

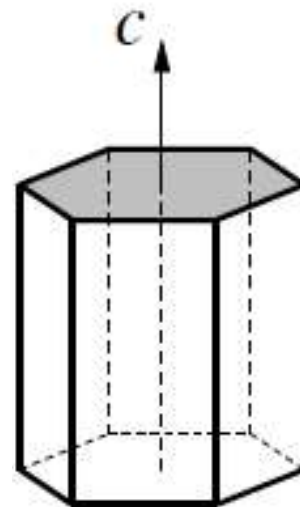
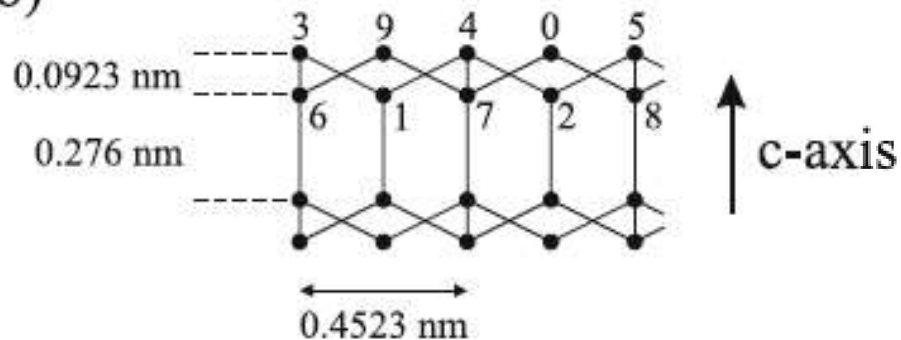
Ice Crystal Structure and Elastic Wave propagation



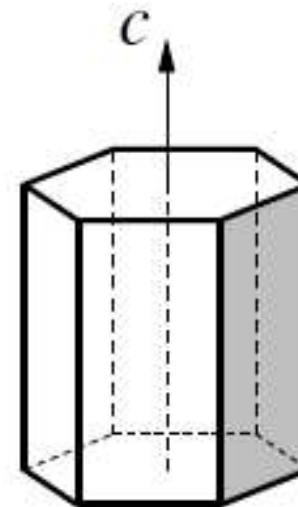
(a)



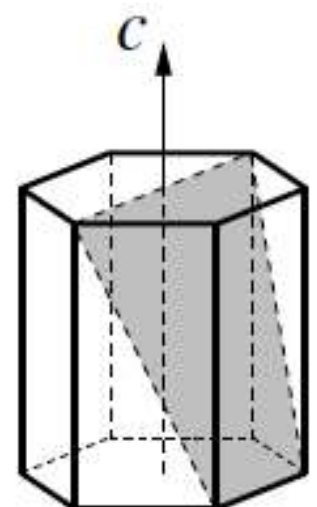
(b)



Basal



Prismatic



Pyramidal

Shear Wave Splitting and Fabric

